

Basic Principles of Medicine II

Module: Nervous System and Behavioral Science | Lecture NB 13

Cranial Nerves V and VII

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Recommended reading

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 5th ed. Elsevier. Chapter 8.



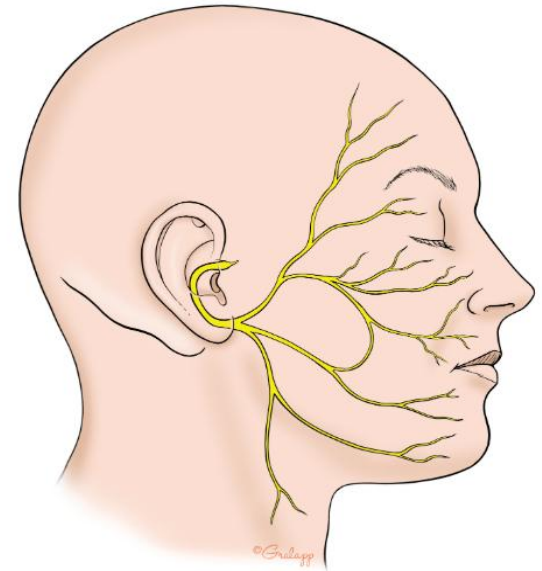
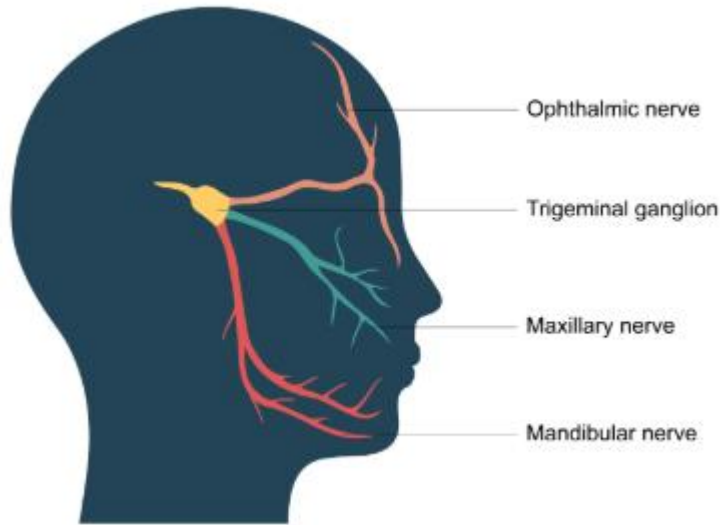
Pay special attention to the 'In the Clinic' boxes which contain the clinical correlates for this topic.



Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0157	Describe how and where the trigeminal nerve (CN V) attaches to the brainstem
SOM.MKII.BPM2.1.NB.1.ANAT.0158	Describe the locations of the three nuclei of the trigeminal nerve and their associated structures
SOM.MKII.BPM2.1.NB.1.ANAT.0159	Describe the three sensory divisions of the trigeminal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0160	Describe the motor component of the trigeminal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0161	Describe the intracranial portion of the trigeminal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0162	Describe the trigeminal ganglion, including its composition and location
SOM.MKII.BPM2.1.NB.1.ANAT.0163	Describe the foramina through which each of the divisions of the trigeminal nerve pass through
SOM.MKII.BPM2.1.NB.1.ANAT.0164	Describe the extracranial pathway of each of the divisions of the trigeminal nerve and the structures that they supply
SOM.MKII.BPM2.1.NB.1.ANAT.0165	Describe the consequence of damage to the trigeminal nerve at different locations: brainstem, in the cavernous sinus, and at each foramen of exit
SOM.MKII.BPM2.1.NB.1.ANAT.0166	Describe how and where the facial nerve attaches to the brainstem
SOM.MKII.BPM2.1.NB.1.ANAT.0167	Describe the locations of the nuclei of the facial nerve and their associated structures
SOM.MKII.BPM2.1.NB.1.ANAT.0168	Describe the different components of the facial nerve and their origin
SOM.MKII.BPM2.1.NB.1.ANAT.0169	Describe the parasympathetic component of the facial nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0170	Describe the intracranial portion of the facial nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0171	Describe the geniculate ganglion, including its composition, location and branches
SOM.MKII.BPM2.1.NB.1.ANAT.0172	Describe the pathway the facial nerve takes to pass from the intra- to extra-cranial portions
SOM.MKII.BPM2.1.NB.1.ANAT.0173	Describe the extracranial pathway of the facial nerve, its branches, and the structures that they supply
SOM.MKII.BPM2.1.NB.1.ANAT.0174	Describe the consequence of damage to the facial nerve at different locations: brainstem, in the facial canal, and at each foramen

Overview



- *Cranial nerves: V, VII*
- *Trigeminal nerve and Facial nerve*
 - *Nuclei*
 - *Ganglia*
 - *Branches*



To understand the cranial nerves discussed in this lecture, know:

- Location, name and type of **nucleus**
- **Attachment** to the brainstem/brain
- Where does it **exit (or enter)** the skull
- **Intracranial/extracranial pathway** (if applicable)
- Structures that it **supplies** along its path and its functional modality
- *What would be the **deficiency** if damaged at each of the above locations*

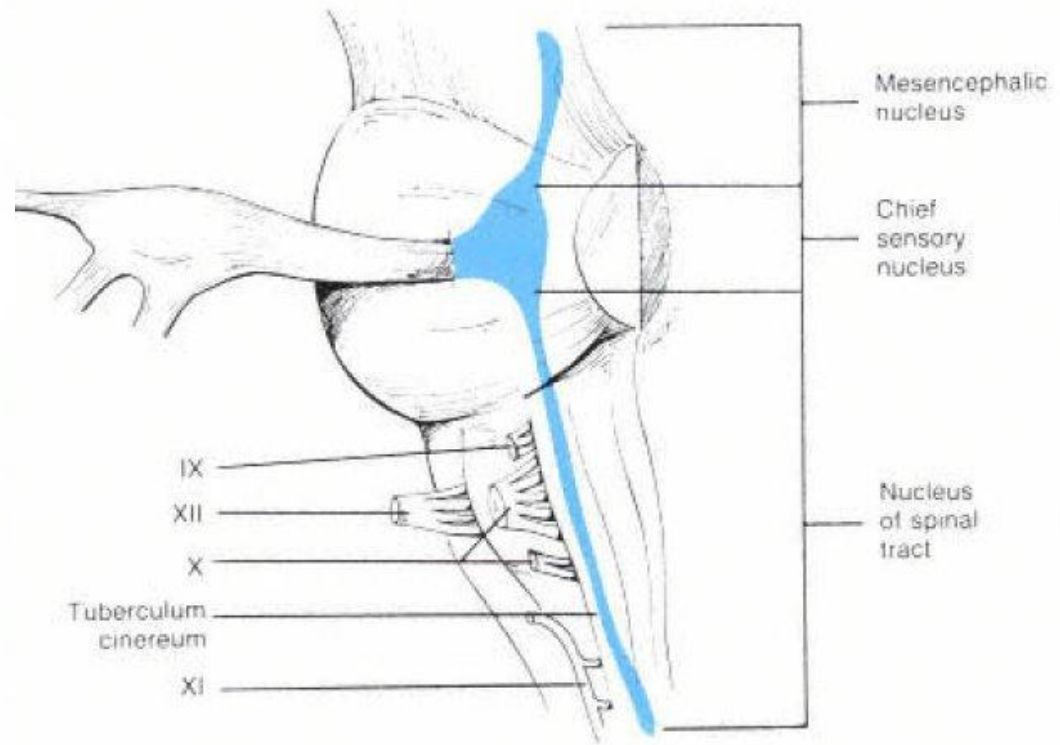
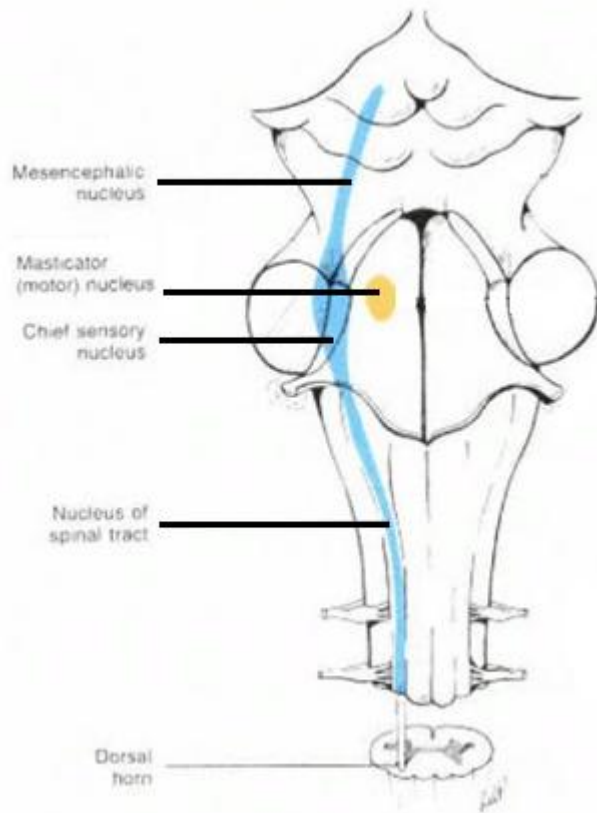


Trigeminal nerve - Overview

Trigeminal Nerve

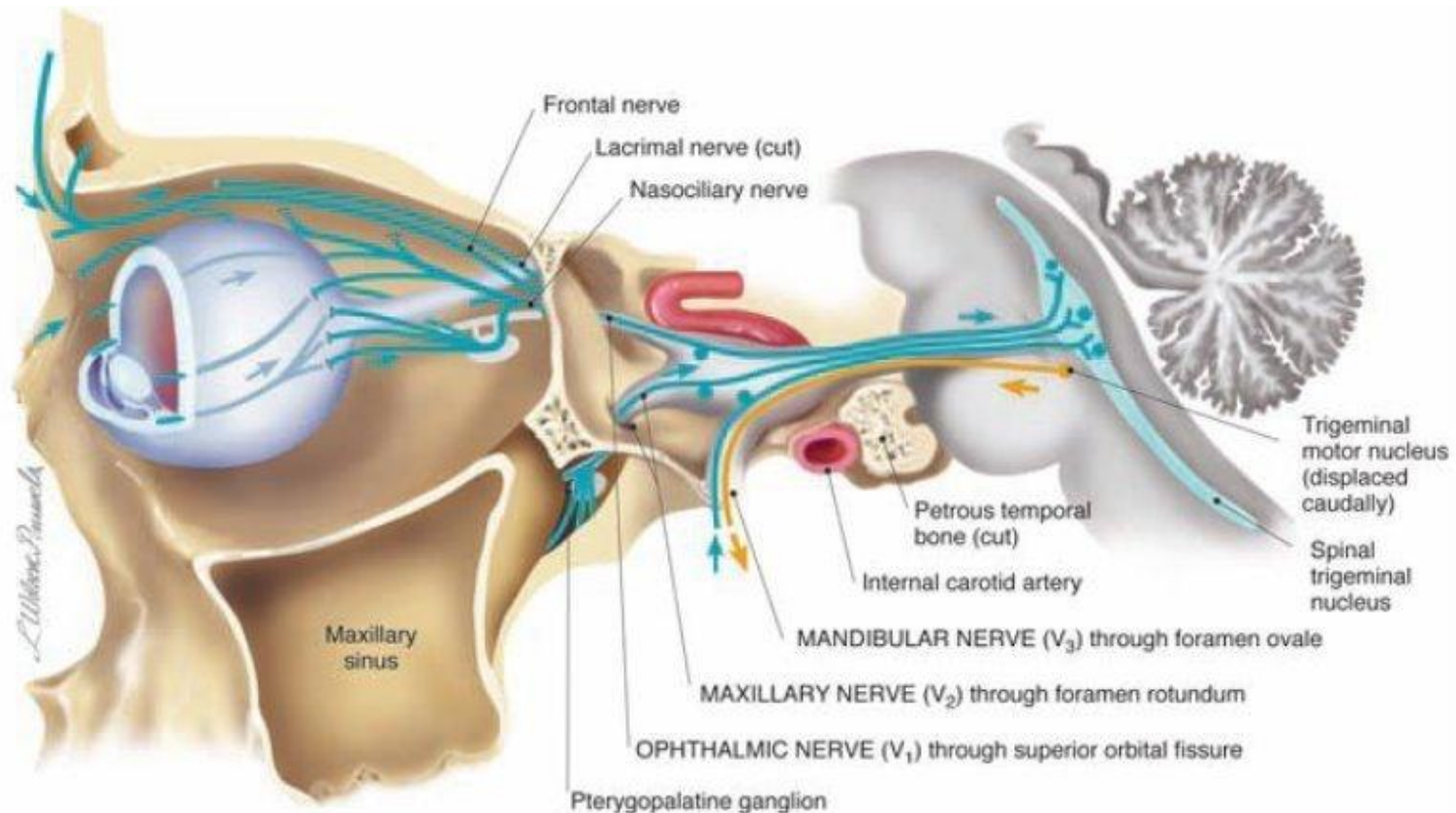
Number	V
Nuclei	3 sensory (chief, mesencephalic, spinal) 1 motor (trigeminal)
Attachment to brainstem	Laterally, merging with middle cerebellar peduncle
Major branches	Ophthalmic (sensory) Maxillary (sensory) Mandibular (sensory AND motor)
Ganglia	Trigeminal in posterior portion of cavernous sinus

Trigeminal nerve - nuclei



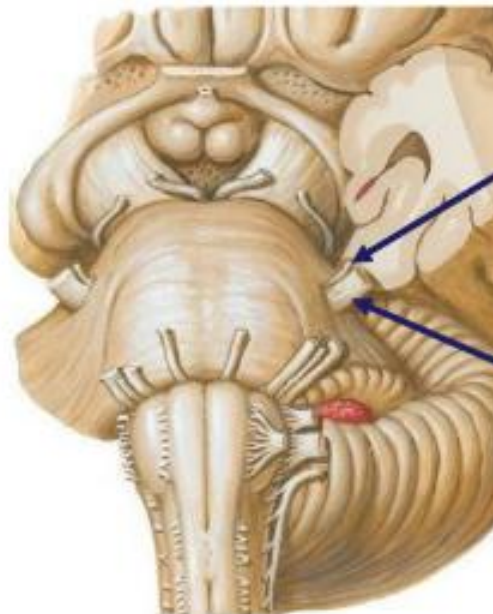
Trigeminal ganglion

- i.e. Gasserian/semilunar ganglion
- Pure **sensory** ganglion
- Found in the middle cranial fossa
 - Located on the lateral wall of the posterior cavernous venous sinus in a depression of the temporal bone (trigeminal/Meckel's cave)



Trigeminal nerve - branching

- Provides sensory to the face and motor to the muscles of first pharyngeal arch
- Divides into 3 major divisions, each serving a “third” of the face
 - CN V_1 , V_2 , V_3



Motor root

Attaches to the lateral portion of the pons

Sensory root

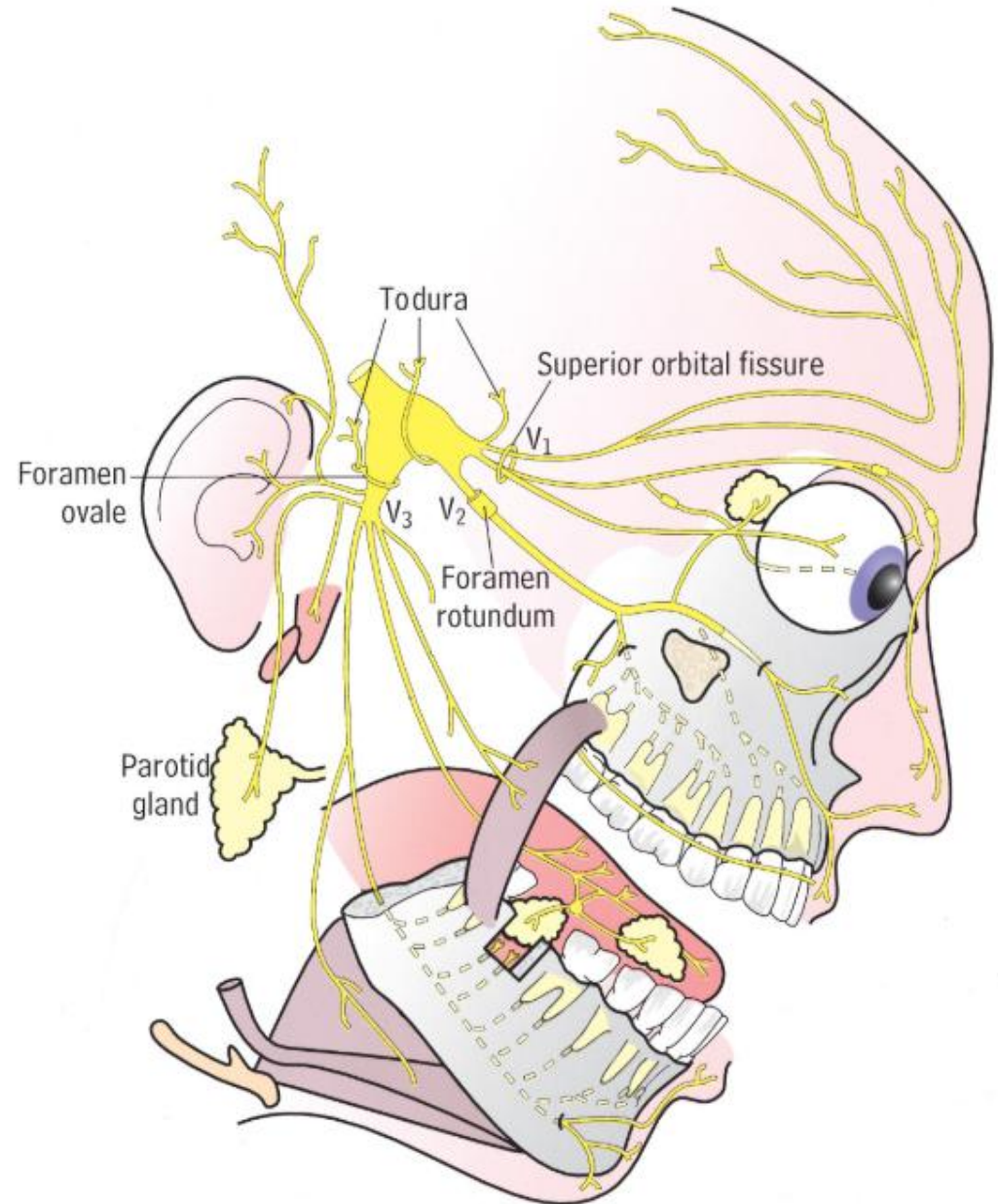
Situated close to the middle cerebellar peduncle

Trigeminal nerve - branches

Ophthalmic (CN V₁): sensory

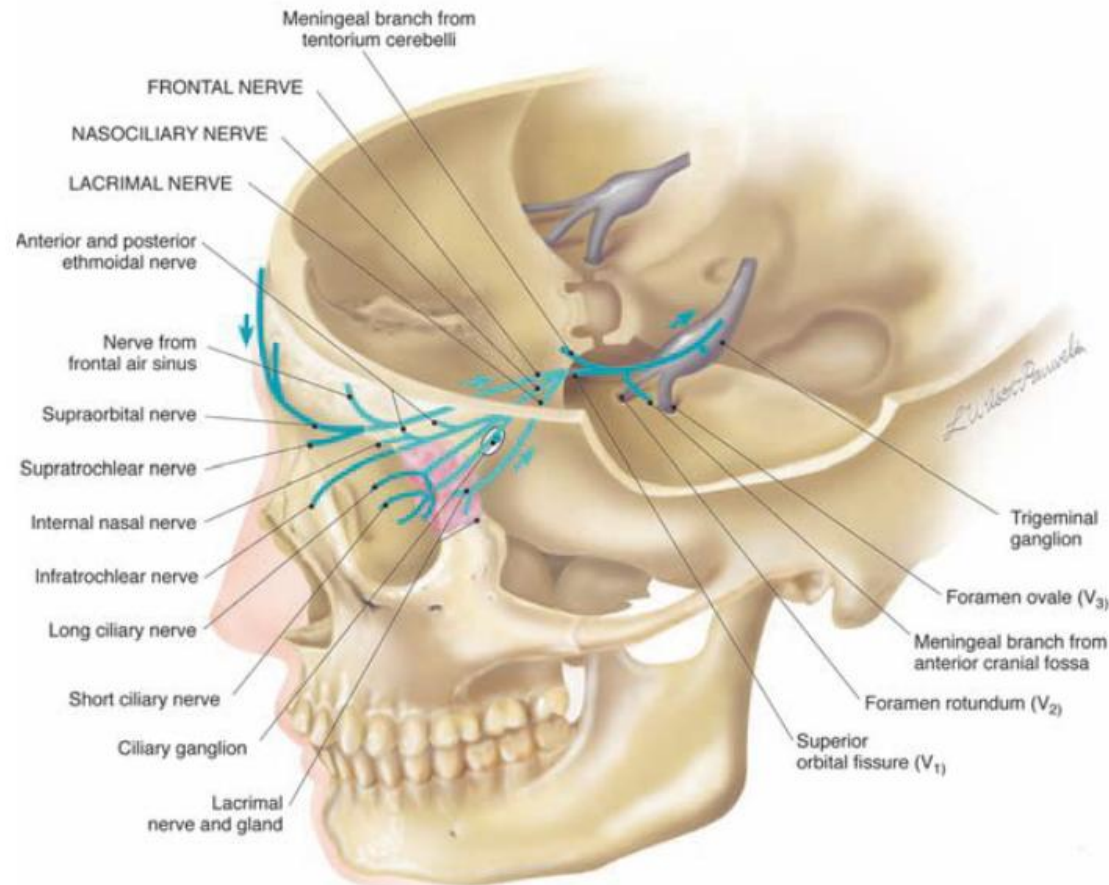
Maxillary (CN V₂): sensory

Mandibular (CN V₃):
sensory **and** motor



Ophthalmic nerve (V_1) - course

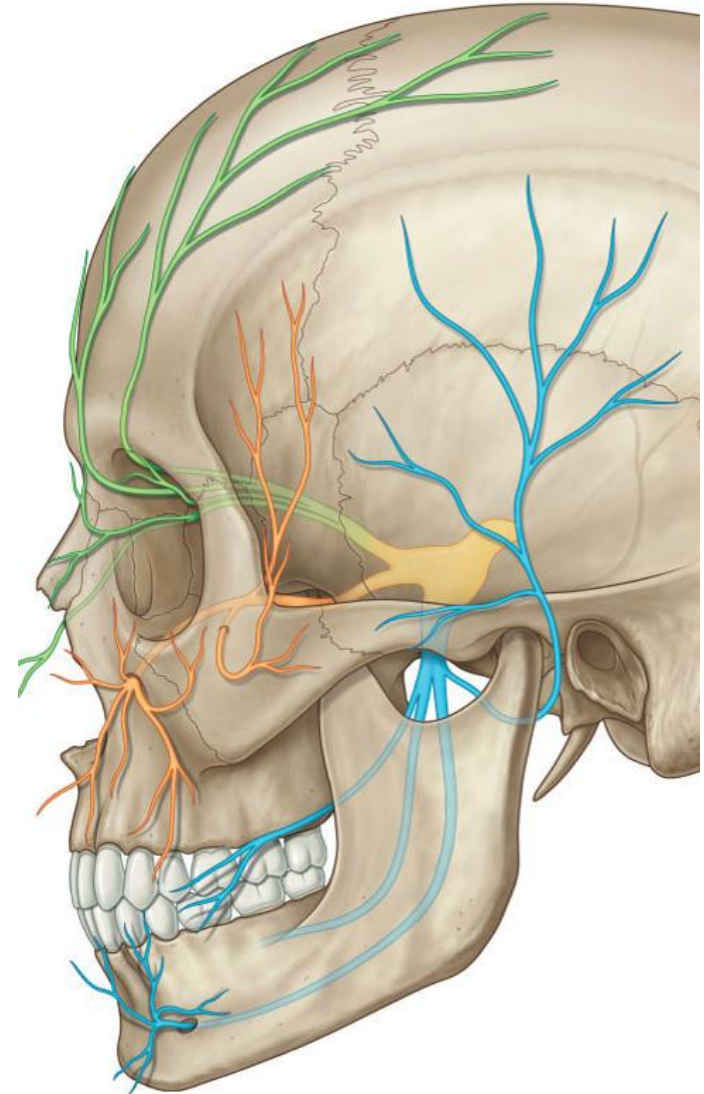
- After arising from the trigeminal ganglion, the ophthalmic nerve travels laterally to the cavernous sinus
 - gives rise to the recurrent tentorial branch (which supplies the tentorium cerebelli).
- Passes through **superior orbital fissure**



Ophthalmic nerve (V_1) - regions supplied

Sensory nerve of the orbit and surrounding area

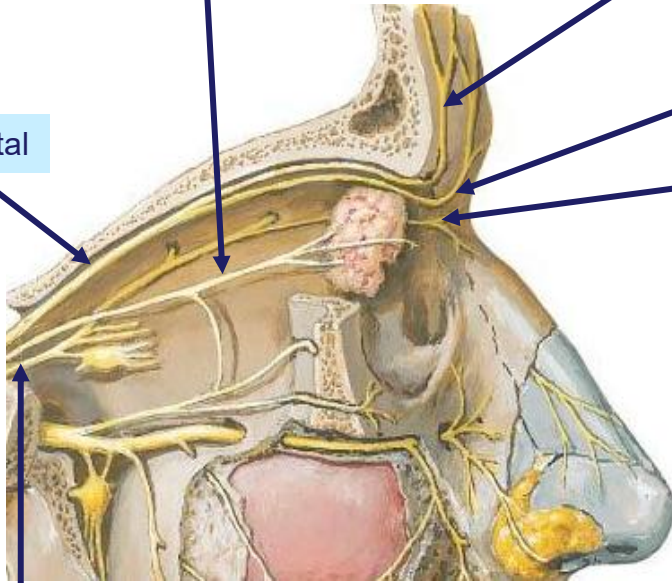
- Meninges (dura mater)
- Forehead
- Orbital region
 - Cornea and eyeball
 - Conjunctiva
 - Lacrimal gland
- Sinuses
 - Ethmoid
 - Frontal
 - Sphenoid
- Nasal region
 - Central portion of bridge of the nose
 - Internal portion of nasal alae
 - Most anterosuperior portion of nasal septum



Ophthalmic nerve (V_1) - branching

Lacrimal - Receives autonomics via connecting branch to zygomatic

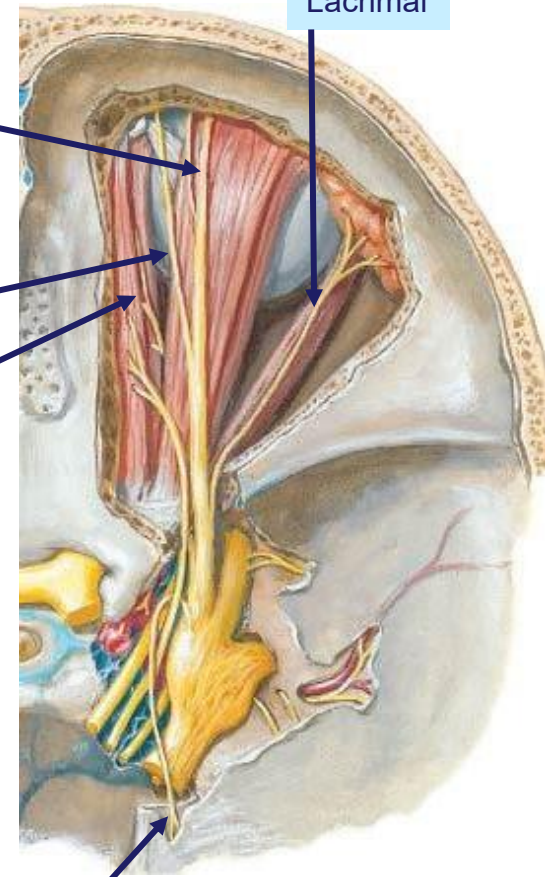
Frontal



Nasociliary

- Gives anterior & posterior ethmoidal branches
- Receives sensory from eyeball
- Mediates corneal reflex

Lacrimal



Dural branch

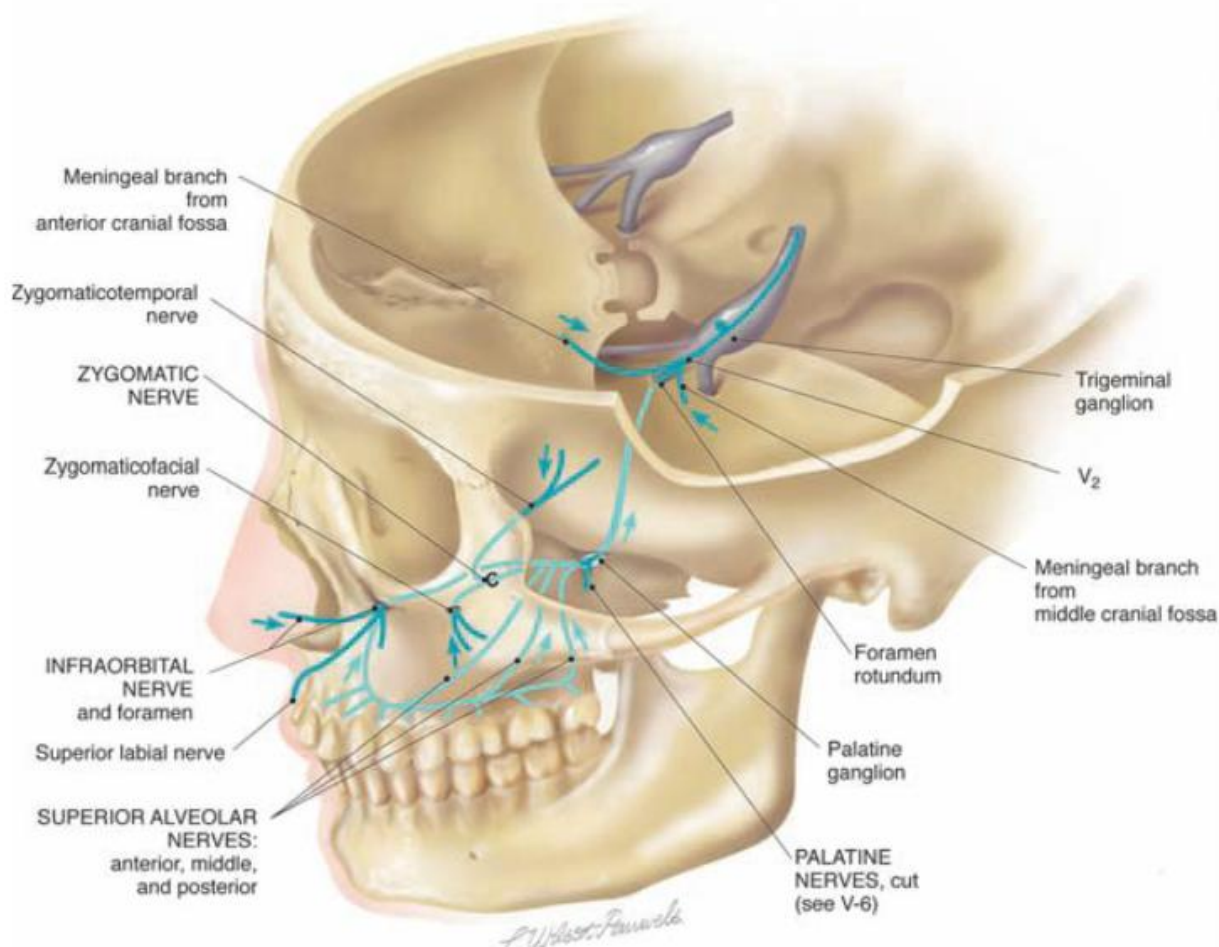


Ophthalmic nerve (V_1) - branches

- After exiting the cranium via the superior orbital fissure, the ophthalmic nerve divides into its **three main branches**:
- **Frontal nerve**
 - **Supraorbital**: innervates upper eyelid, conjunctiva and scalp
 - **Supratrochlear**: innervates upper eyelid, conjunctiva and forehead
- **Lacrimal nerve**
 - Supplies lateral half of upper eyelid and receives branch from zygomatic nerve of CN V_2 containing parasympathetic fibers
- **Nasociliary nerve**
 - **Anterior ethmoid nerve**: Sensory innervation of mucous membranes of frontal, ethmoid, sphenoid sinuses and nasal cavity
 - **Posterior ethmoid nerve**: Sensory innervation to mucous membranes of sphenoid sinus
 - **Infra-trochlear nerve**: Bridge of nose, upper eyelid and conjunctiva
 - **Long ciliary nerves**: Sensory innervation to eye (cornea, ciliary bodies, iris)
 - Contains sympathetic fibers to dilator pupillae muscles

Maxillary nerve (V₂) - course

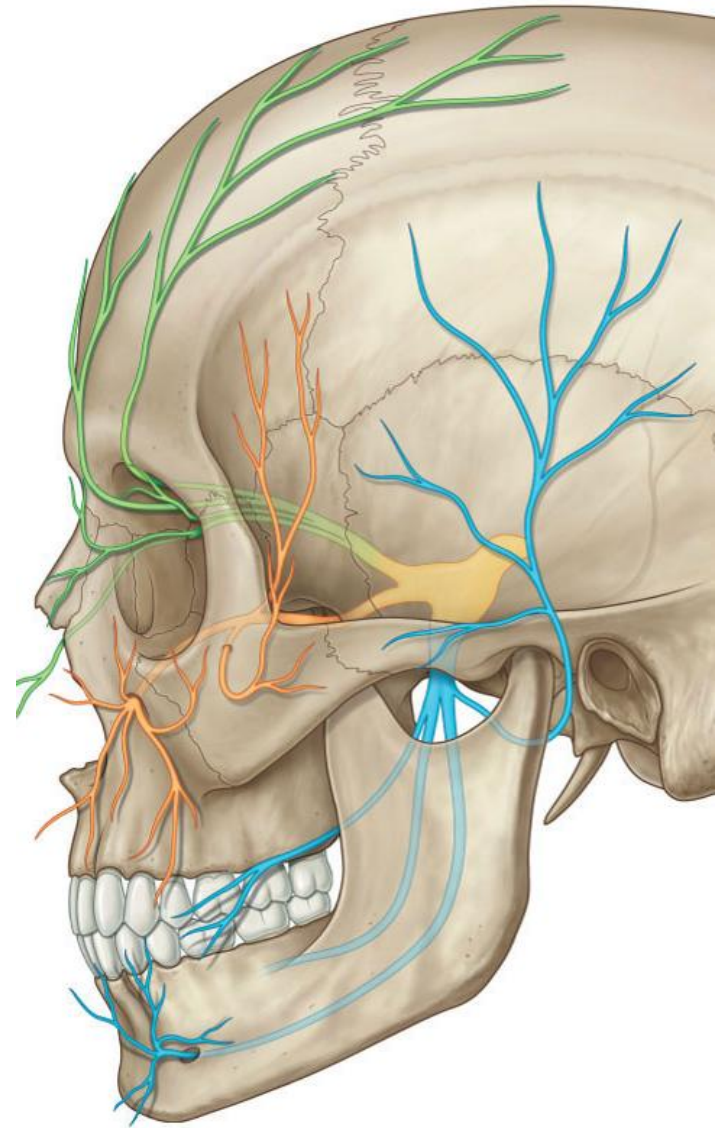
- After arising from the trigeminal ganglion, the maxillary nerve passes through the lateral wall of the cavernous sinus
- Leaves the skull through the **foramen rotundum**
- Continues into the **infraorbital foramen** and exits to the face



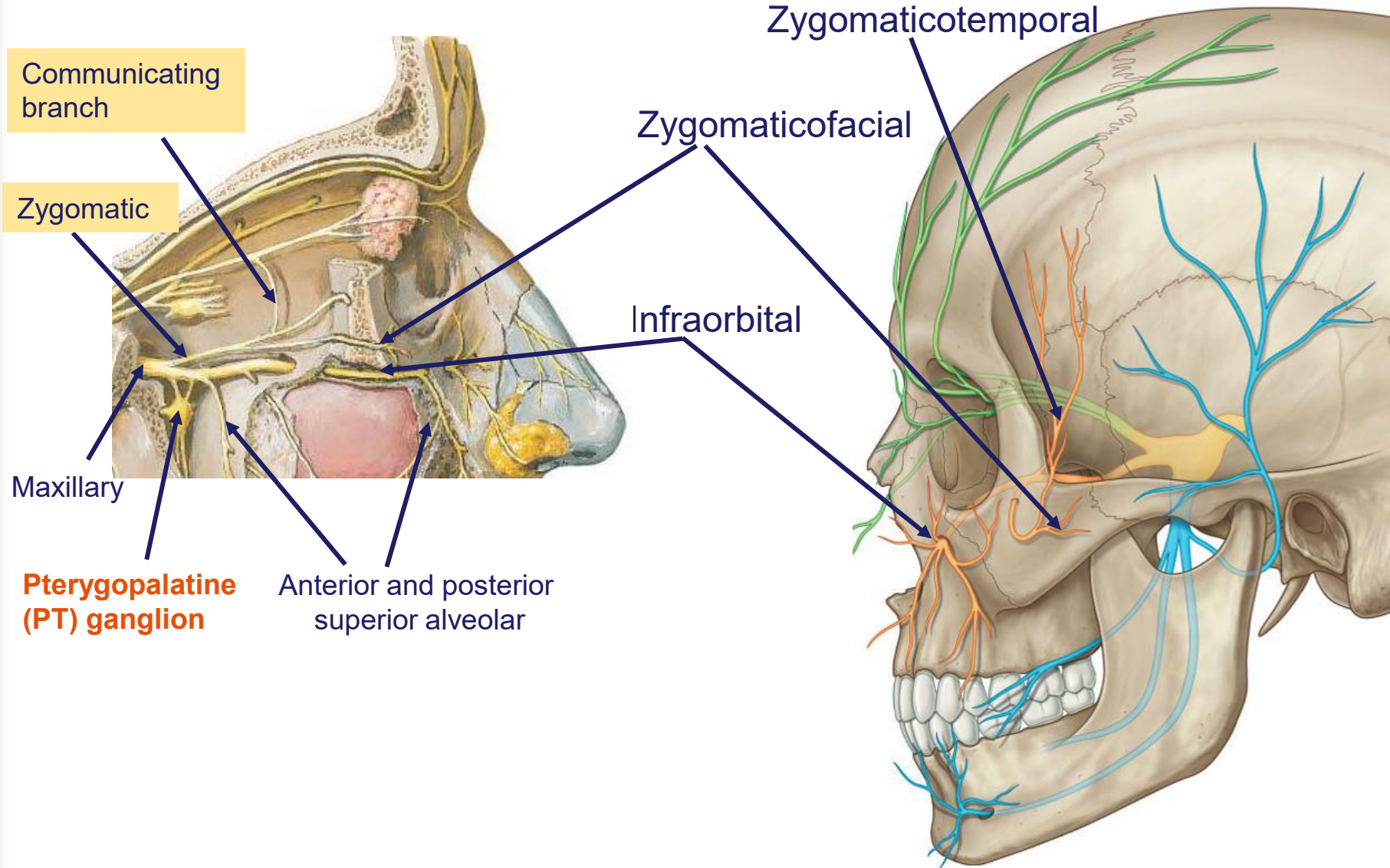
Maxillary nerve (V_2) – regions supplied

Sensory nerve for pterygopalatine fossa, nose, and surrounding area

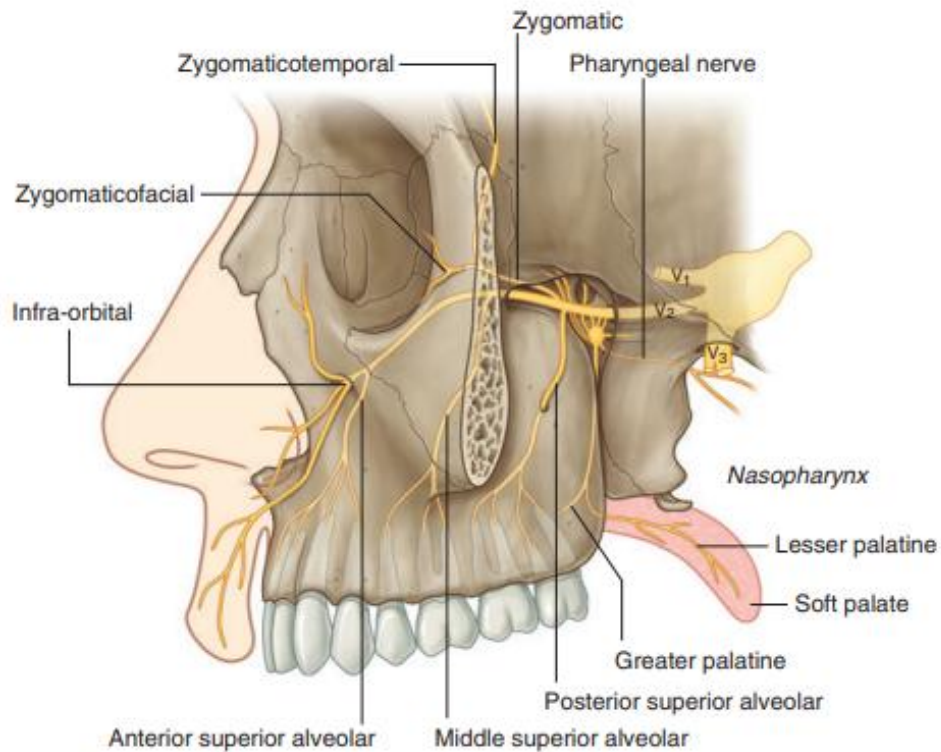
- Meninges (dura mater)
- Lateral temple
- Lower eyelid
- Maxillary sinus
- Cheek
- Upper lip, upper jaw teeth



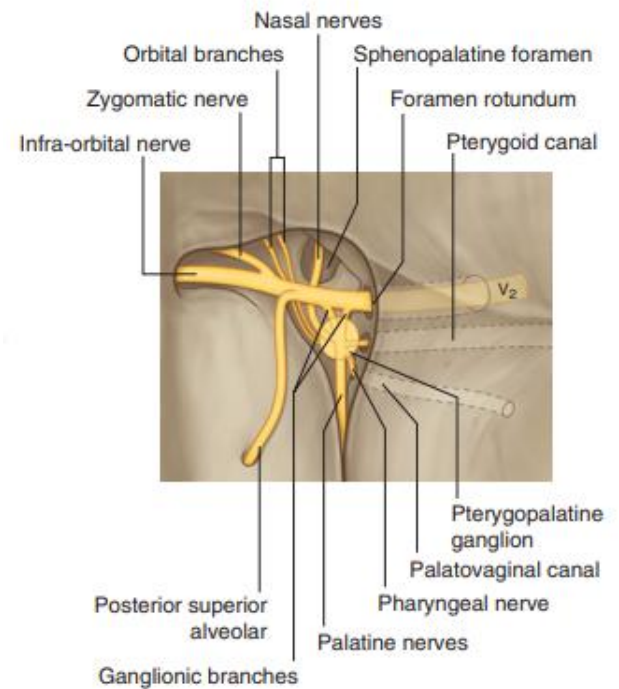
Maxillary nerve (V_2) – branching



PT ganglion



- | | |
|---------------------------------------|-------|
| Preganglionic parasympathetic nerves | — |
| Postganglionic parasympathetic nerves | |
| Preganglionic sympathetic nerves | — |
| Postganglionic sympathetic nerves | |



Nerve of pterygoid canal



Maxillary nerve (V₂) – periorbital branches

Zygomatic nerve

- Enters orbit through **inferior orbital fissure**

Branches into:

Zygomatico-temporal nerve (via zygomaticotemporal foramen)

- sensory innervation of the scalp over anterior half of the temporal fossa above the zygomatic arch
- carries parasympathetic secretomotor fibers to the lacrimal gland via the **lacrimal nerve**.

Zygomatico-facial nerve (via zygomaticofacial foramen)

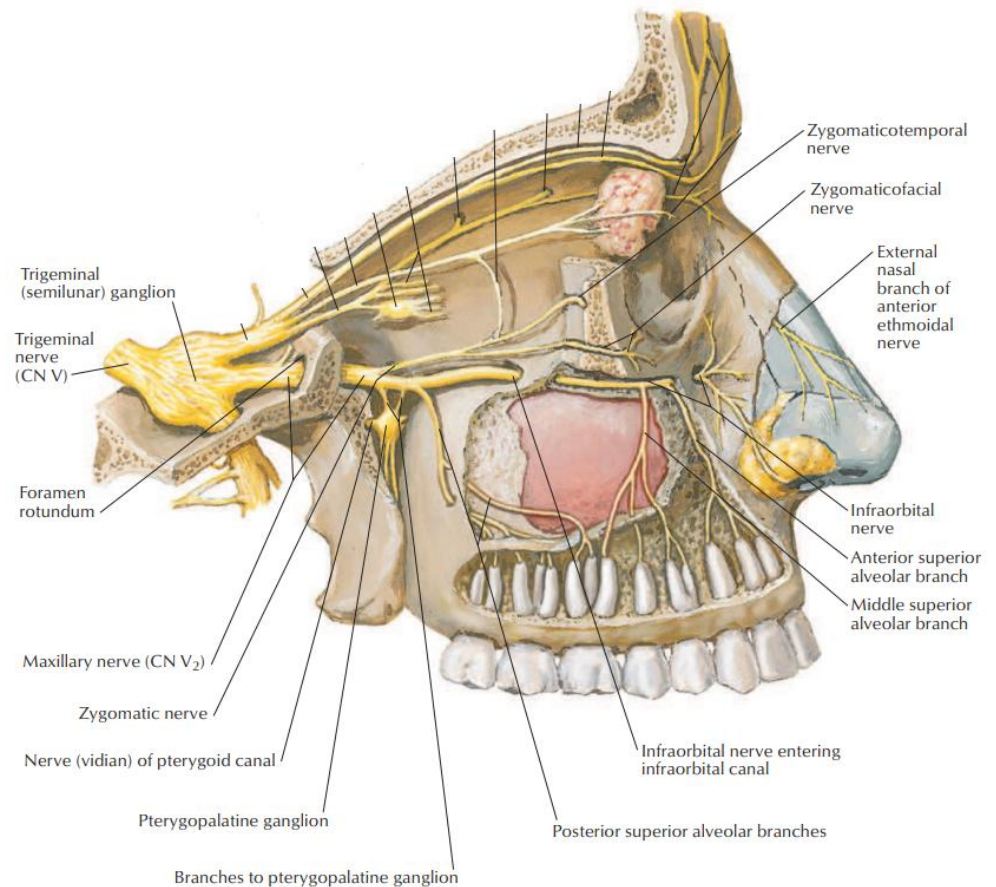
- sensory innervation of scalp over the zygomatic arch and the upper cheek

Infraorbital nerve

- Enters through **inferior orbital canal**

Branches: Anterior and Middle superior alveolar nerves

- Sensory innervation to the superior dental plexus
- Then passes through **infraorbital foramen** (located in maxilla bone)
 - Sensory innervation to medial cheek, lateral nose, mucosa of the anteroinferior nasal septum, upper lip



Maxillary nerve (V₂) – nasal and oral cavity branches

Nasopalatine nerve

- Enters nasal cavity through **sphenopalatine foramen**
- Sensory innervation of the mucosa of posteroinferior nasal septum

Pharyngeal nerve

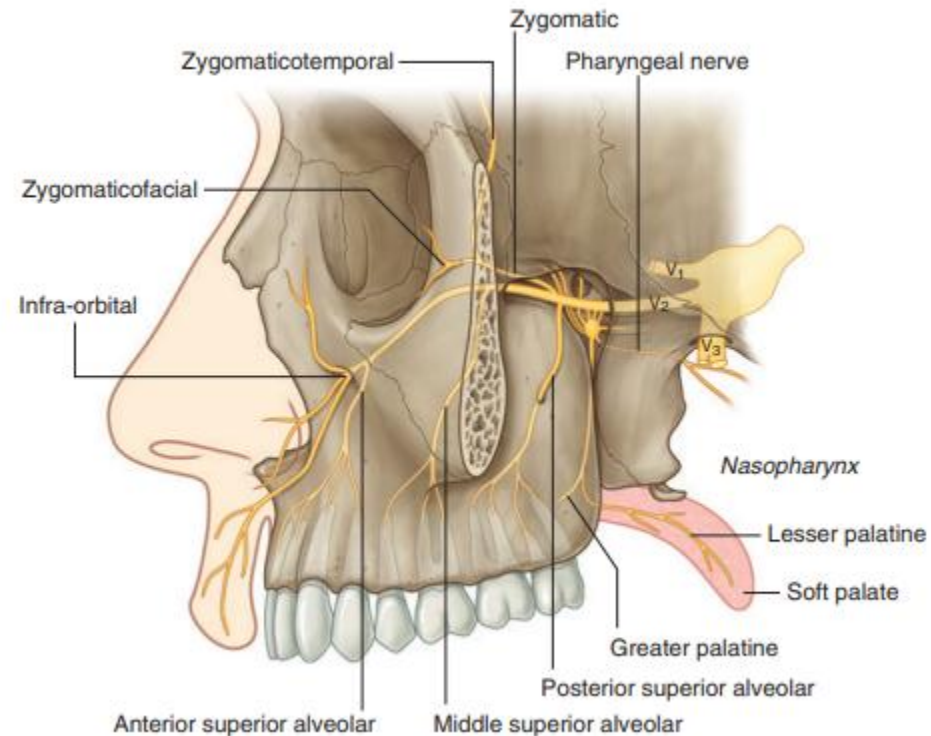
- Enters nasal cavity through **palatovaginal canal**
- Sensory innervation of mucosa of the superior nasopharynx

Greater palatine nerve

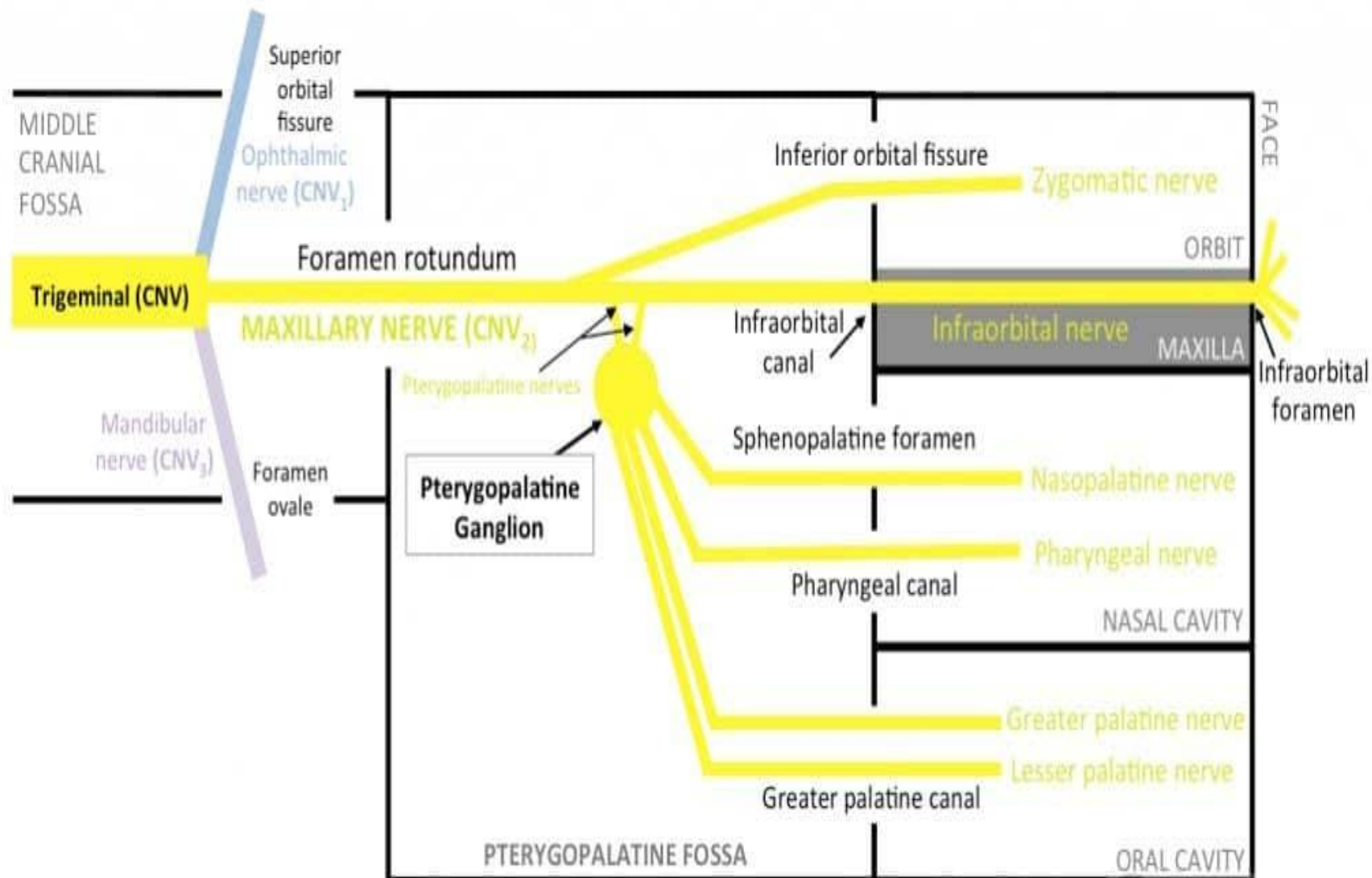
- Enters oral cavity through **greater palatine foramen**
- Sensory innervation of the mucosa of the posterior hard palate

Lesser palatine nerve

- Enters oral cavity through **lesser palatine foramen**
- Sensory innervation of mucosa of the uvula, soft palate, and palatine tonsils

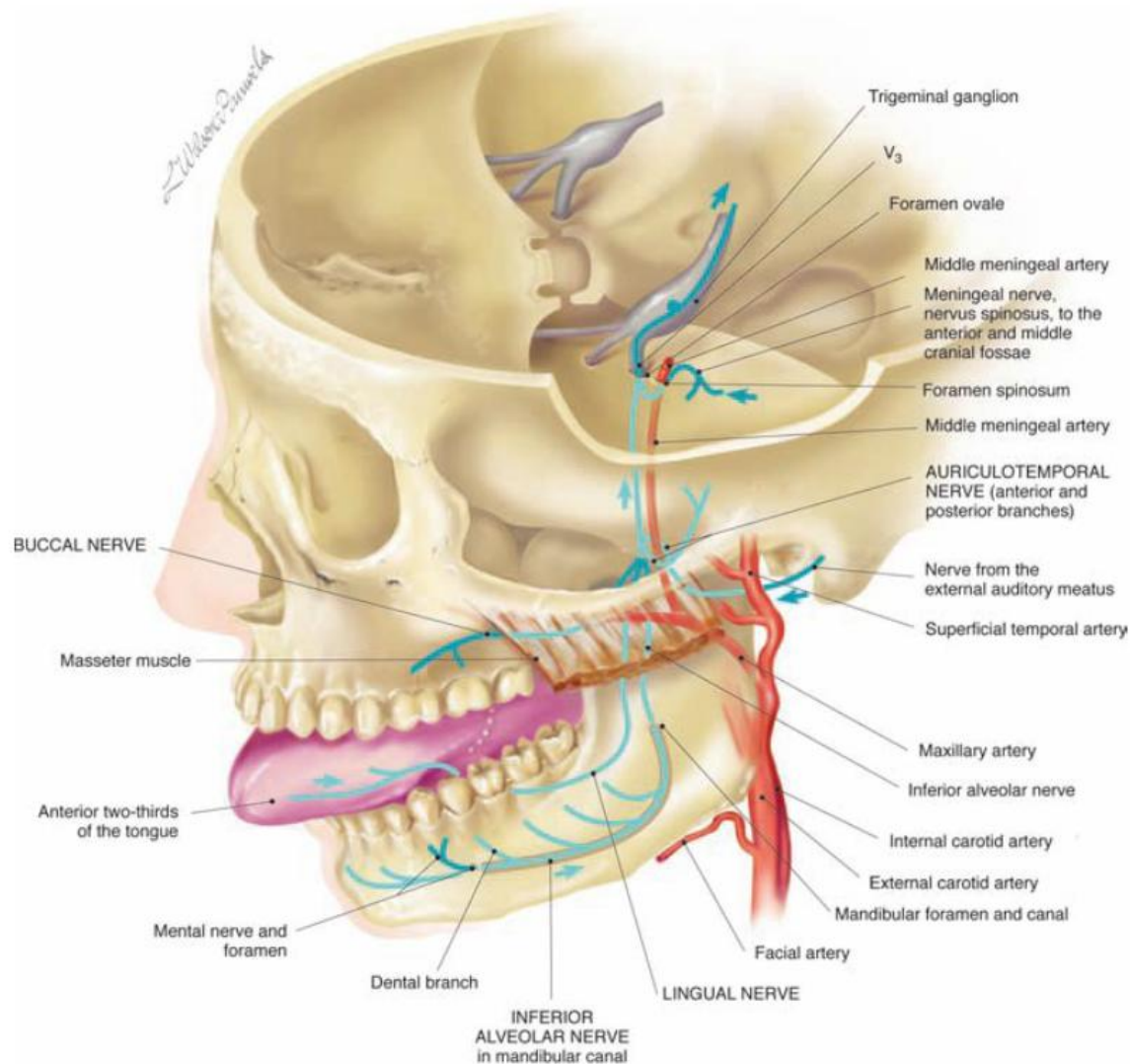


Maxillary nerve (V₂) – summary



Mandibular nerve (V_3) - course

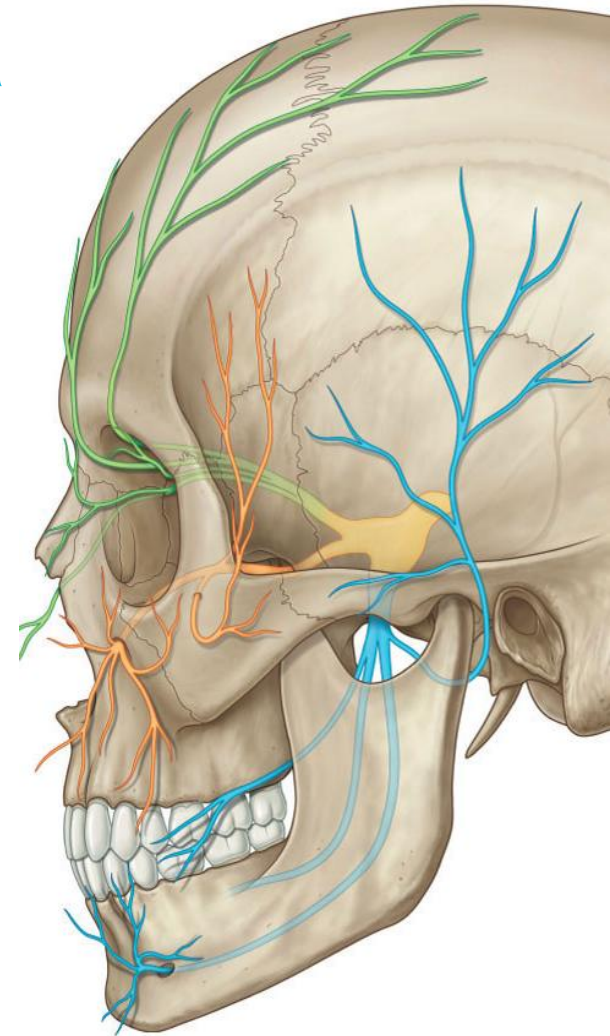
- Passes through **foramen ovale** into the **infratemporal fossa**
- Does **NOT** pass through the cavernous sinus



Mandibular nerve (V₃) – sensory regions

Sensory nerve of the oral cavity,
parotid gland and surrounding area

- Meninges (dura mater)
- Tympanic membrane
- Anterior to ear and parotid gland
- Temporomandibular joint
- Mucosa and skin of cheek and chin
- Anterior two-thirds of tongue
- Lower lip, lower jaw teeth

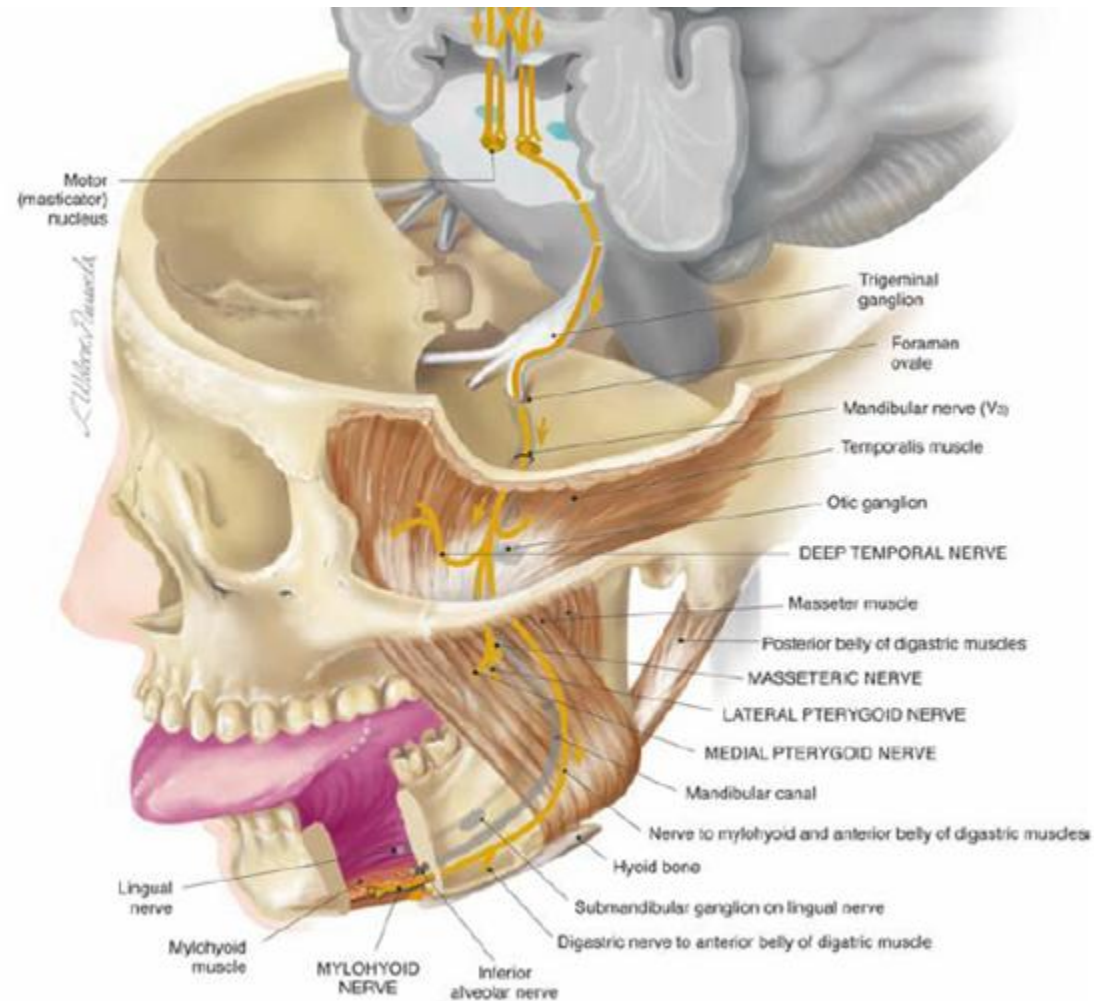


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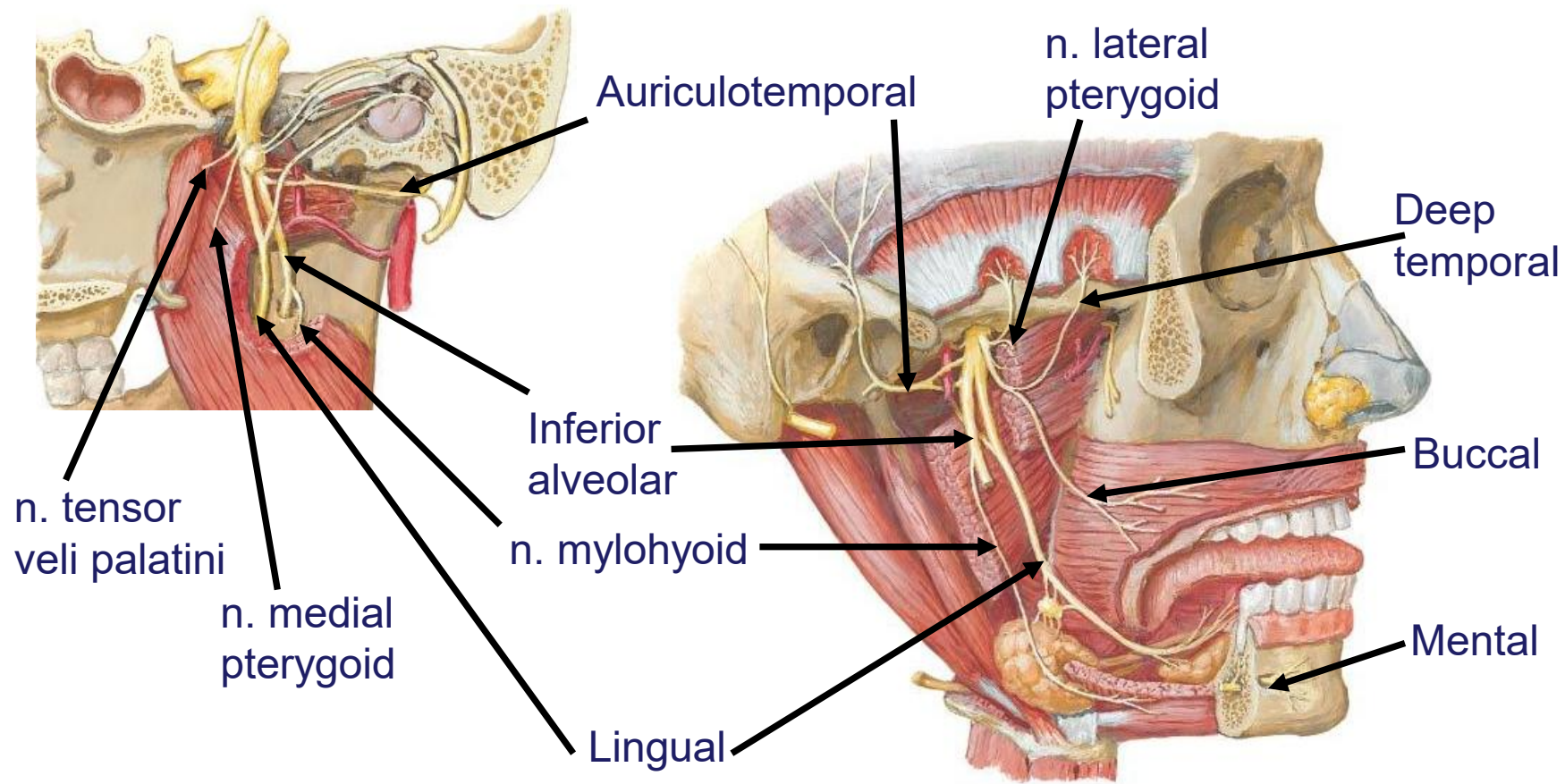
Mandibular nerve (V_3) – motor regions

Motor nerve of the muscles of mastication

- Temporalis
- Medial pterygoid
- Lateral pterygoid
- Masseter
- Anterior belly of digastric
- Tensor veli palatini
- Tensor tympani
- Mylohyoid



Mandibular nerve (V_3) – branching



Mandibular (V₃) - motor branches

- Muscles of **first pharyngeal arch** (MAST MATT)

- Masseter
- Temporalis
- 2 pterygoid muscles (medial and lateral)



Muscles of
Mastication

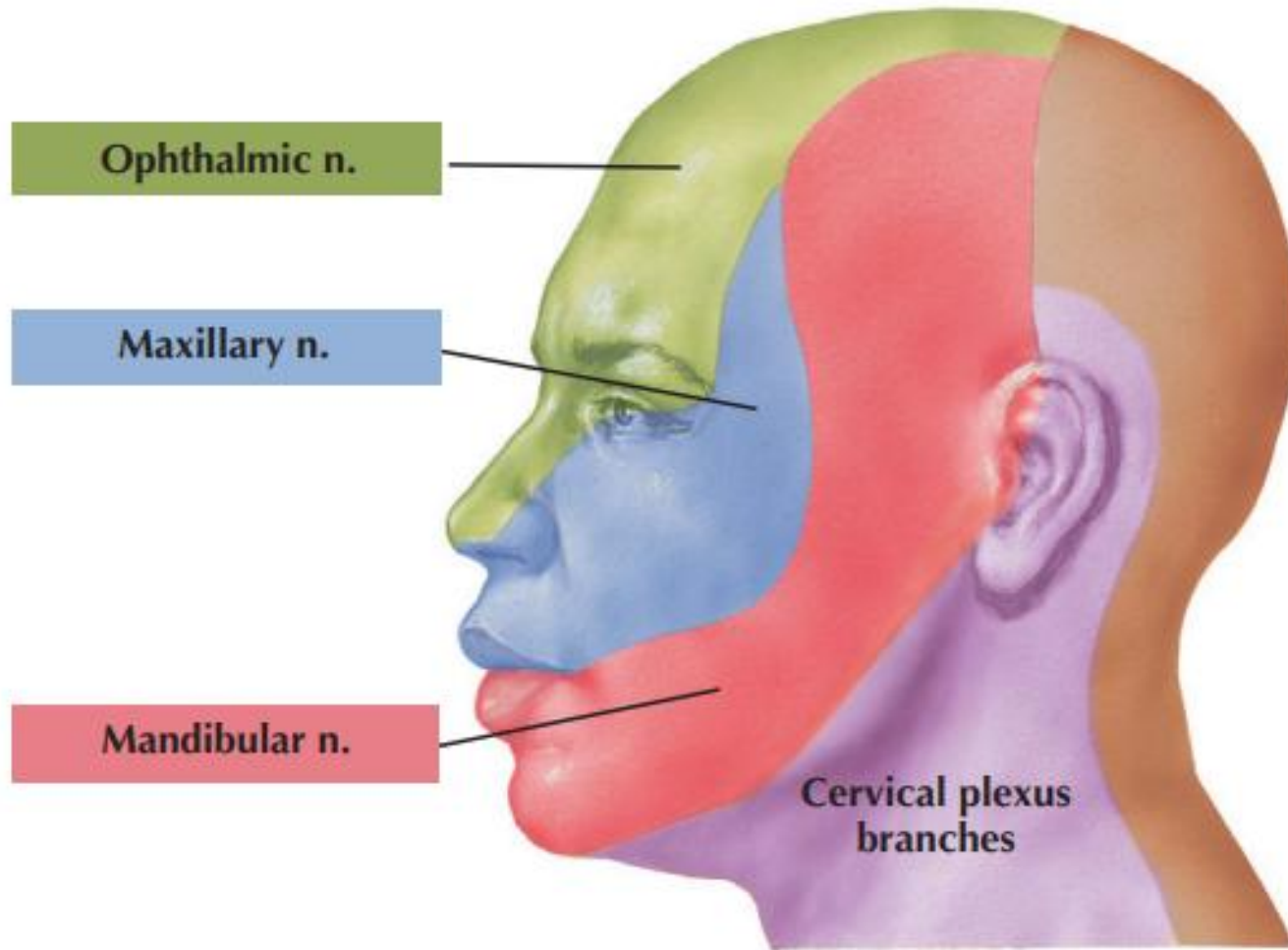
- **Mylohyoid**
 - elevates hyoid bone during swallowing
- **Anterior belly of digastric**
 - elevates hyoid bone during swallowing
- **Tensor veli palatini**
 - elevate the soft palate to prevent regurgitation into the nasopharynx
- **Tensor tympani**
 - dampens sound, such as those created by chewing



Mandibular (V_3) - sensory branches

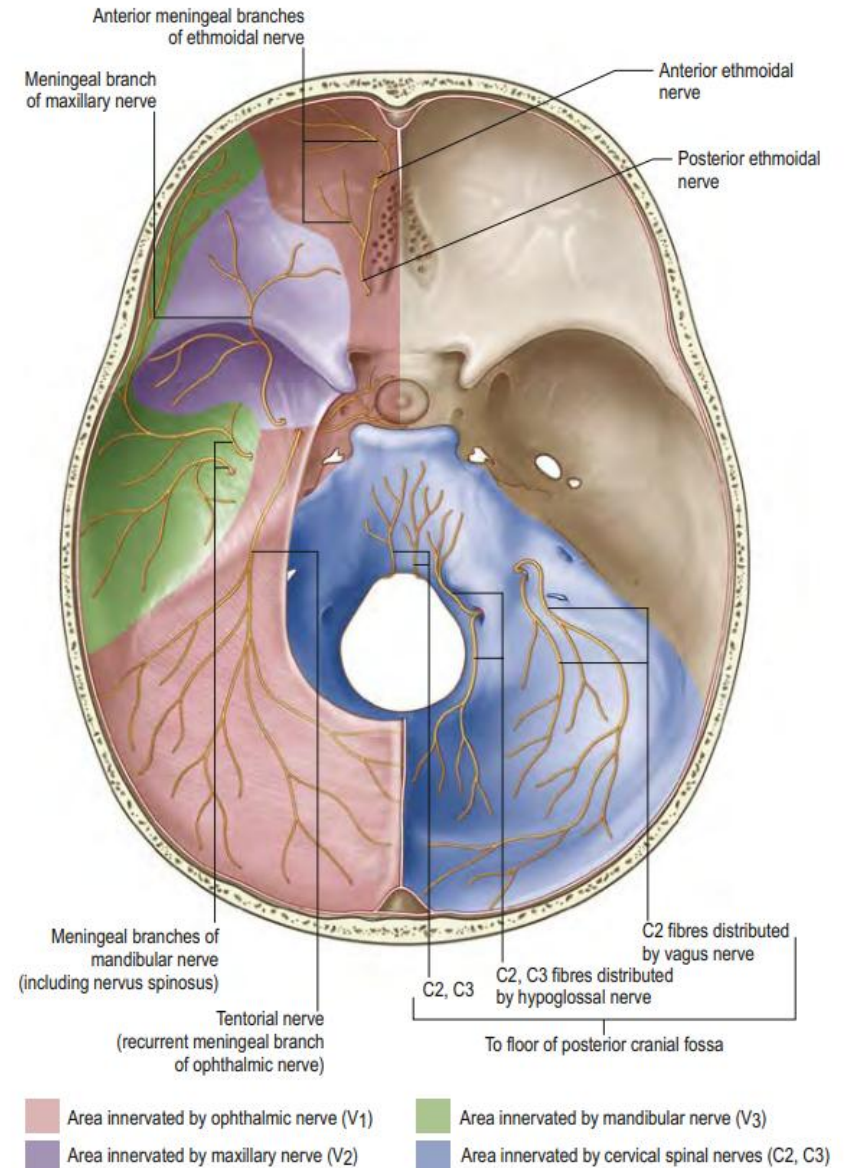
- **Auriculotemporal Nerve**: sensory innervation to
 - Anterior part of the auricle and parotid area
 - Lateral part of the temple
 - Anterior external meatus
 - Anterior tympanic membrane
- **Buccal Nerve**: sensory innervation to
 - buccal membranes of the mouth
 - second and third molar teeth
- **Inferior Alveolar Nerve** (sensory and motor)
 - **Mylohyoid nerve**
 - motor to the mylohyoid and anterior digastric muscles
 - **Mental nerve**
 - sensory
- **Lingual Nerve** (general sensory and special sensory)
 - General sensory fibers: innervate anterior two-thirds of the tongue (V_3)
 - Special sensory fibers: carried with lingual nerve
 - provide taste to the anterior two-thirds of the tongue (VII).

Cutaneous sensory distribution



Intracranial sensory distribution

- Branches of the trigeminal nerve provide innervation to the meninges
- **V₁**: falx cerebri, anterior cranial fossa and tentorium cerebelli
- **V₂**: medial aspect of middle cranial fossa
- **V₃**: lateral aspect of the middle cranial fossa
- Posterior cranial fossa is innervated by the cervical nerves



Clinical Tests for trigeminal nerve

1) Inspection

2) Palpation

- muscles of mastication

3) Sensory evaluation

- touch areas of V_1 , V_2 , V_3

4) Jaw-jerk reflex

- Tests 2 components of V_3 with reflex hammer
 - Sensory branch
 - Motor branch

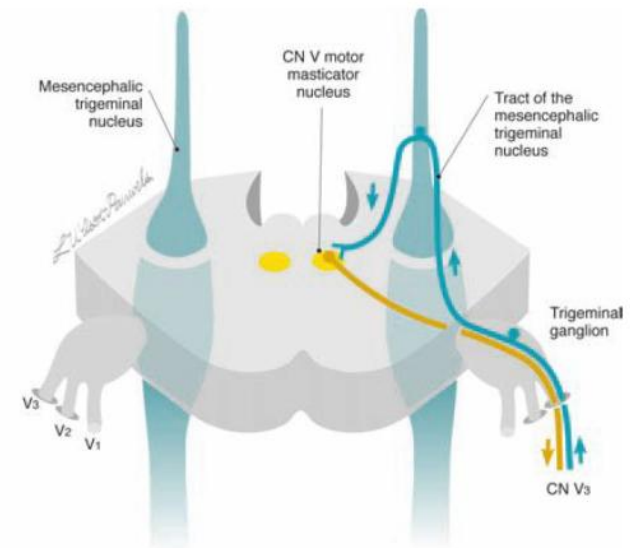
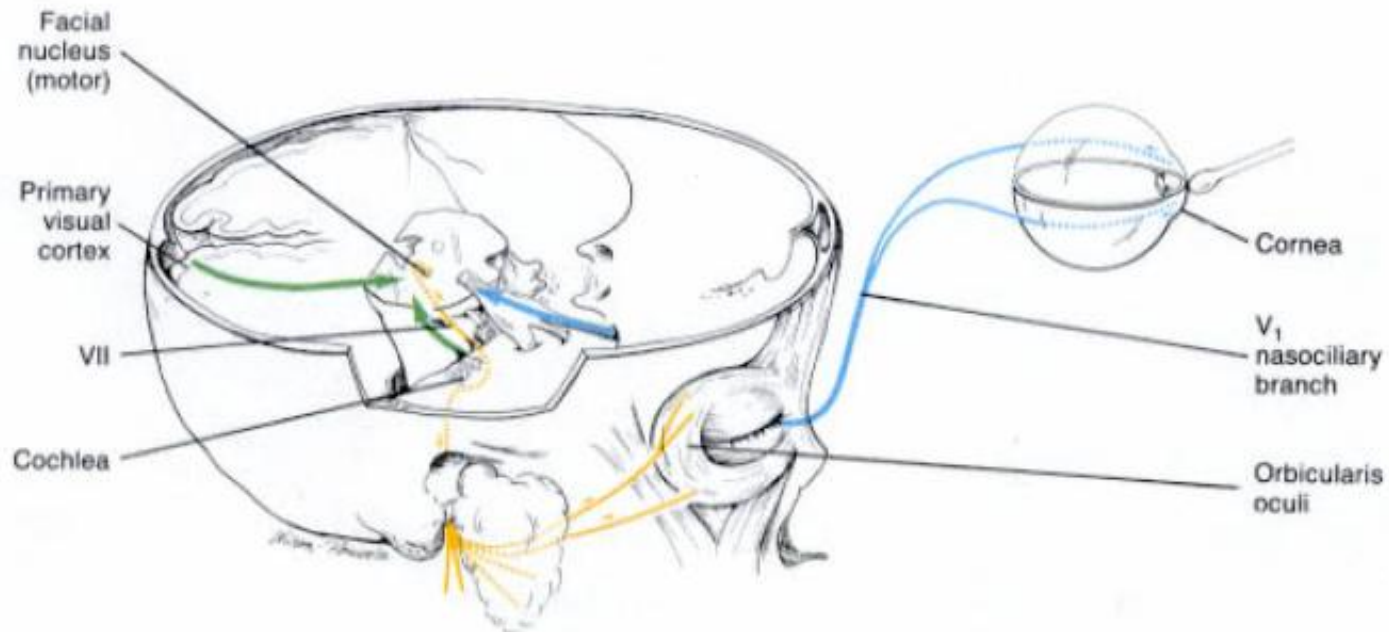


FIGURE V-11 Mandibular jaw reflex (sensory input V_3 , motor output to muscles of mastication V_3).

Clinical Tests for trigeminal nerve

5) Corneal reflex

- aka: blink reflex, eyelid reflex
- touching the cornea leads to eye closure
- consists of 2 components
 - **V₁** : cornea of eye (sensory)
 - **VII**: closes eye (motor)



Clinical Correlate: Shingles

- Herpes zoster virus remains dormant in the cell bodies of ganglia
 - e.g. trigeminal ganglion
- Reactivated viruses follow the dermatomes innervated by the affected (trigeminal) nerve
- Consequences may include
 - vesicular rash
 - blindness
 - toothache
 - neuralgia
 - encephalitis
- Commonly unilateral



Figure 1 Hemifacial vesicular eruption affecting the left maxillary, nasal and periorbital regions.

doi: [10.1136/bcr-2014-208673](https://doi.org/10.1136/bcr-2014-208673)

Facial nerve - overview

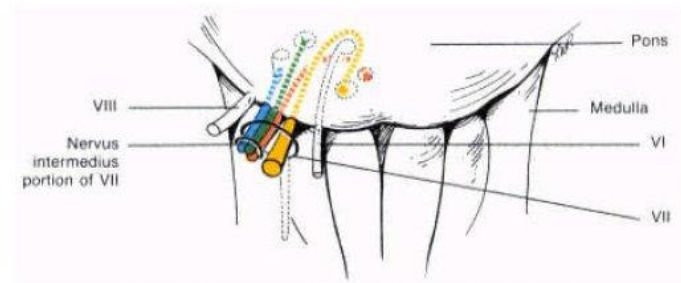
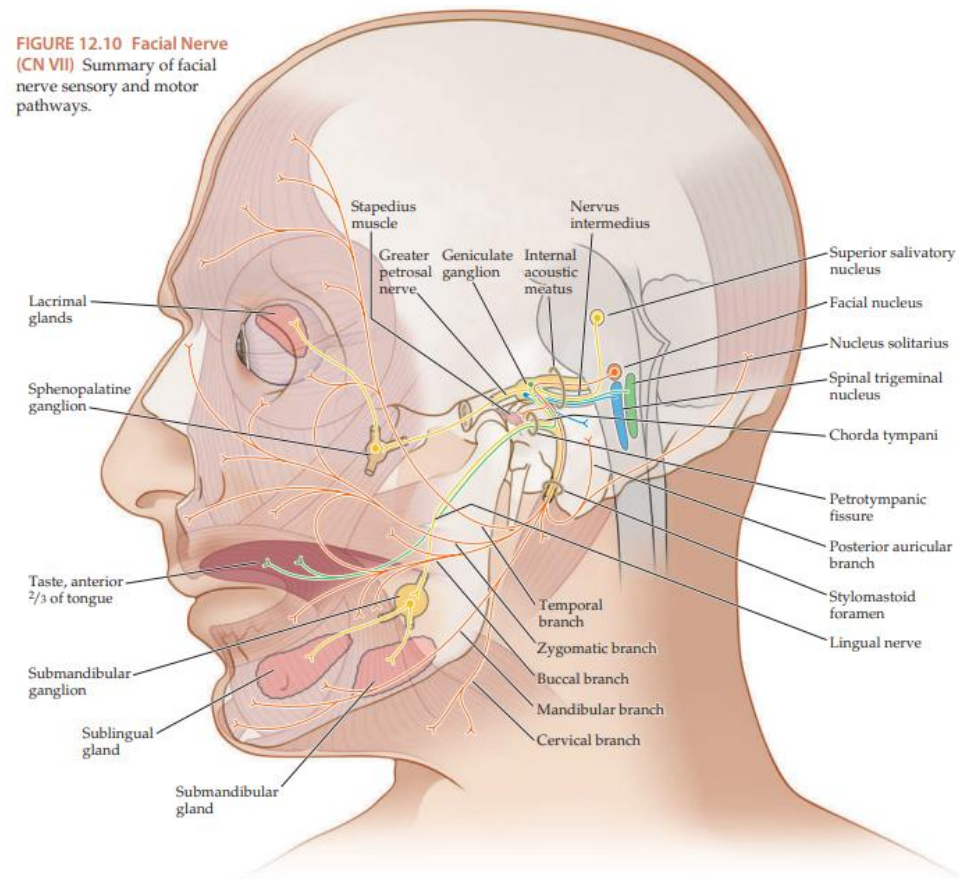
Facial Nerve

Number	VII
Nuclei	1 special sensory (nucleus solitarius) 1 motor (facial) 2 parasympathetic (superior salivatory nucleus, rostral lacrimal nucleus)
Attachment to brainstem	Ventrolateral to the pontomedullary junction
Major branches	Greater petrosal (parasympathetic) Chorda tympani (special sense and parasympathetic) Motor trunk (main portion will branch further)
Ganglia	Geniculate in petrous temporal bone

Facial nerve – nuclei

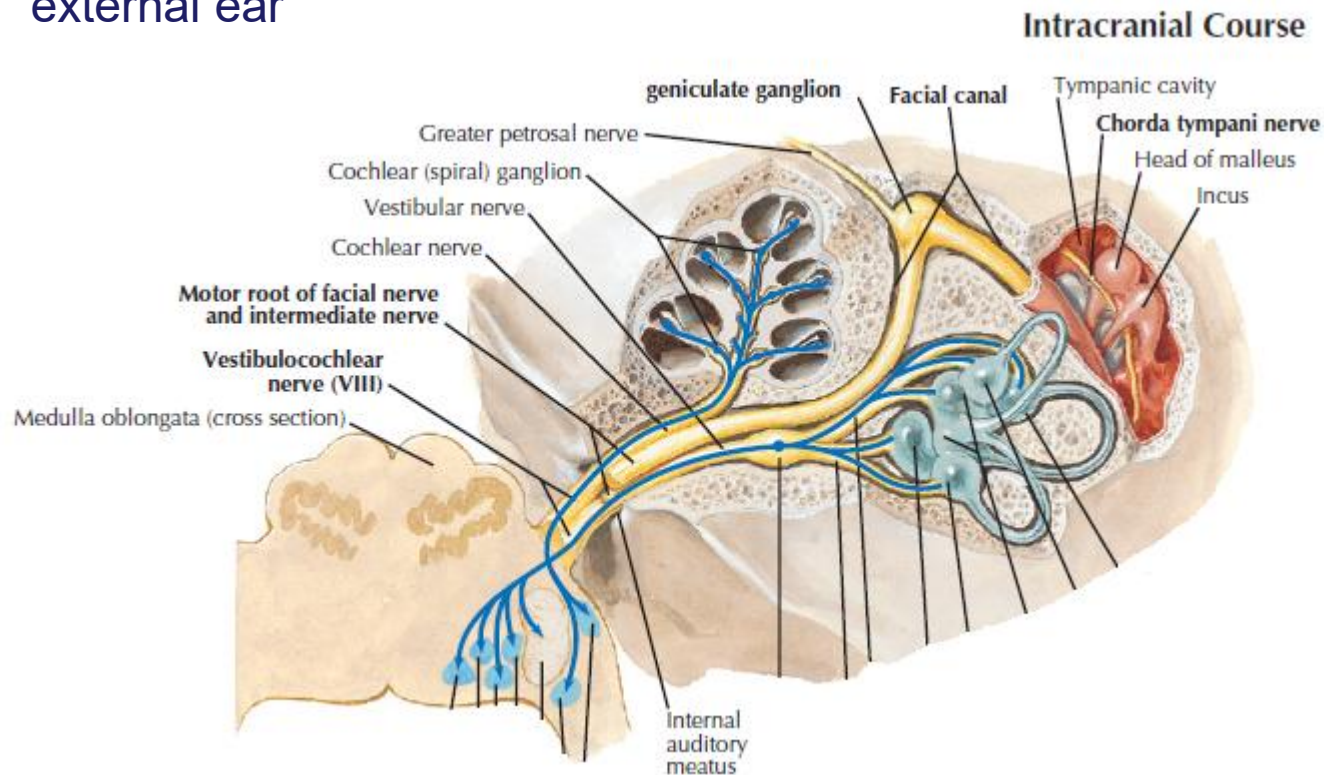
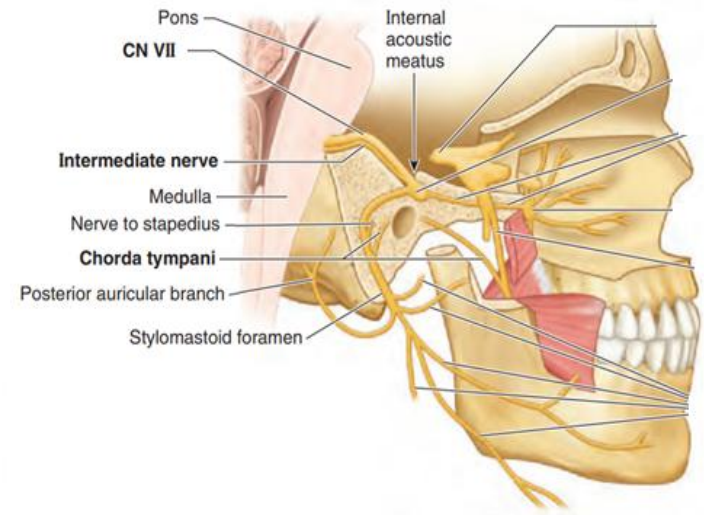
- Motor root
 - Attaches to the ventrolateral aspect of the pontomedullary junction
 - exits the cranium via **internal acoustic meatus**
- Parasympathetic fibers
 - Arise from **superior salivatory nucleus** and **lacrimal nucleus**
 - exit cranium via **internal acoustic meatus**
- Sensory root
 - **nucleus solitarius** in medulla oblongata
 - situated close to VIII
- Special sensory (taste)
 - fibers travel with **nervus intermedius** to brainstem

FIGURE 12.10 Facial Nerve (CN VII) Summary of facial nerve sensory and motor pathways.



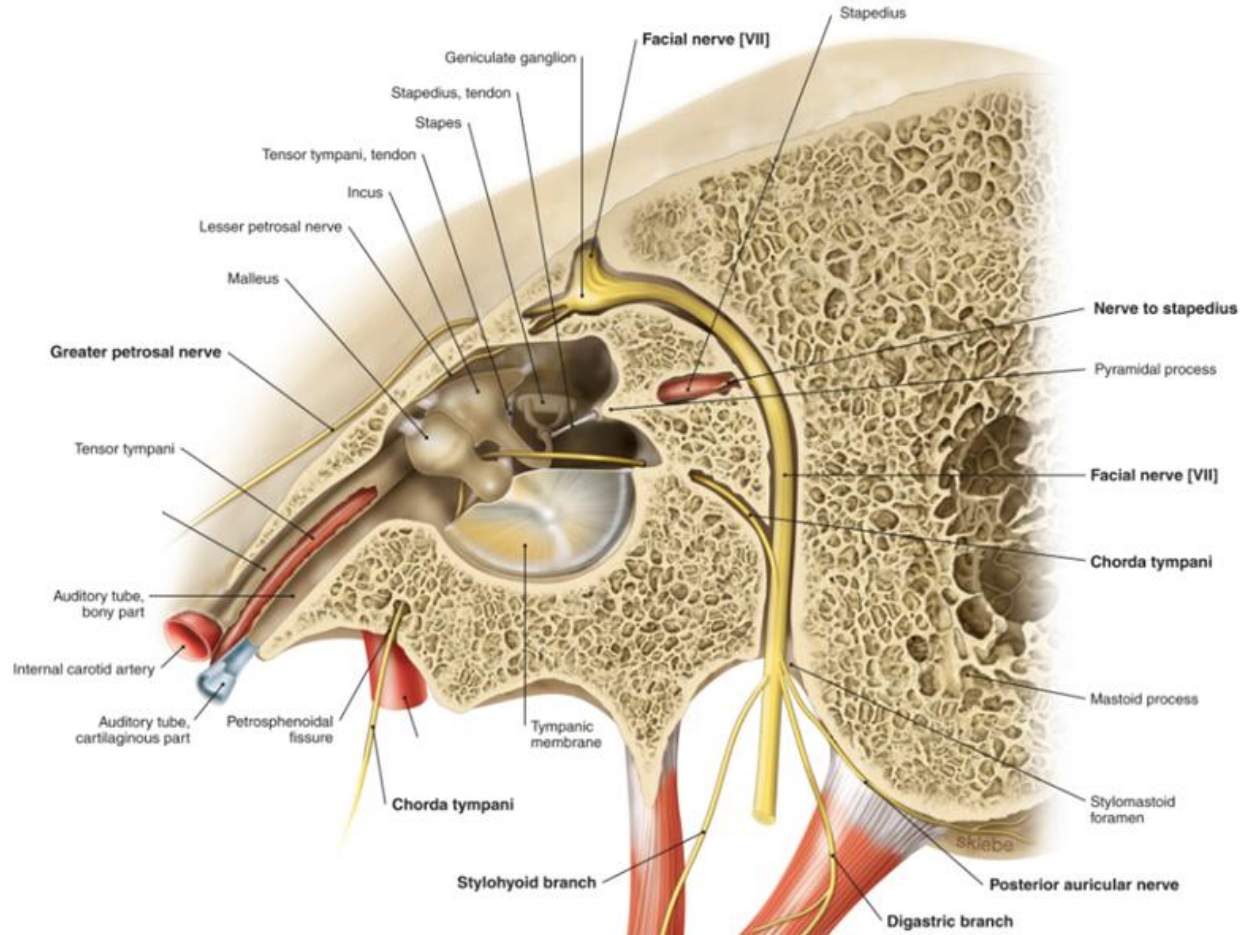
Facial nerve – ganglia

- General sensory
 - Nerve cell bodies within **geniculate ganglion** in petrous temporal bone
 - Exits cranium via **internal acoustic meatus**
 - Cutaneous nerve endings around external ear

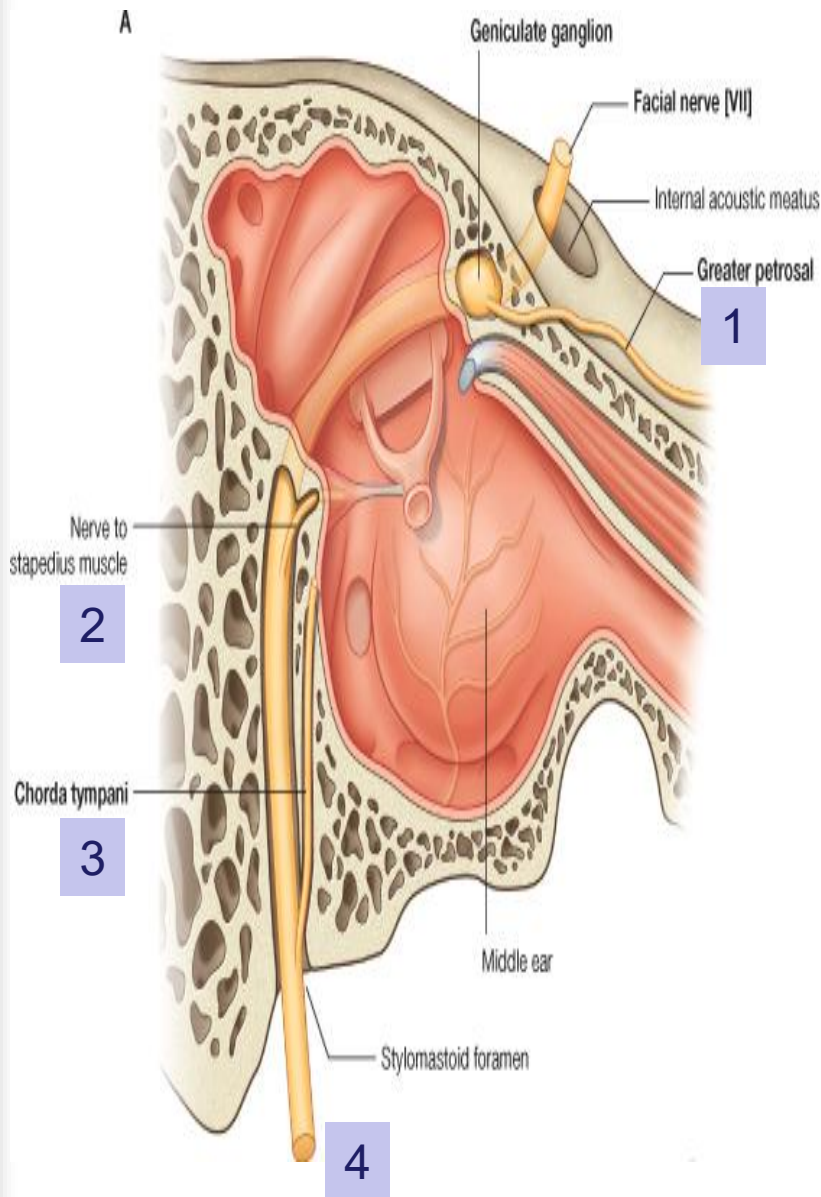


Facial nerve – intracranial course

- Passes through **internal acoustic meatus**
 - Forms *geniculate ganglion*
 - Gives off **greater petrosal nerve**
- Passes through **facial canal**
 - Gives off *nerve to stapedius*
 - Gives off **chorda tympani** as it continues
- Exits **stylomastoid foramen**



Facial nerve - intracranial branches



1. Greater petrosal nerve

Carries *pre-ganglionic parasympathetic fibers* to pterygopalatine ganglion

2. Nerve to stapedius muscle

innervates the stapedius muscle

3. Chorda tympani

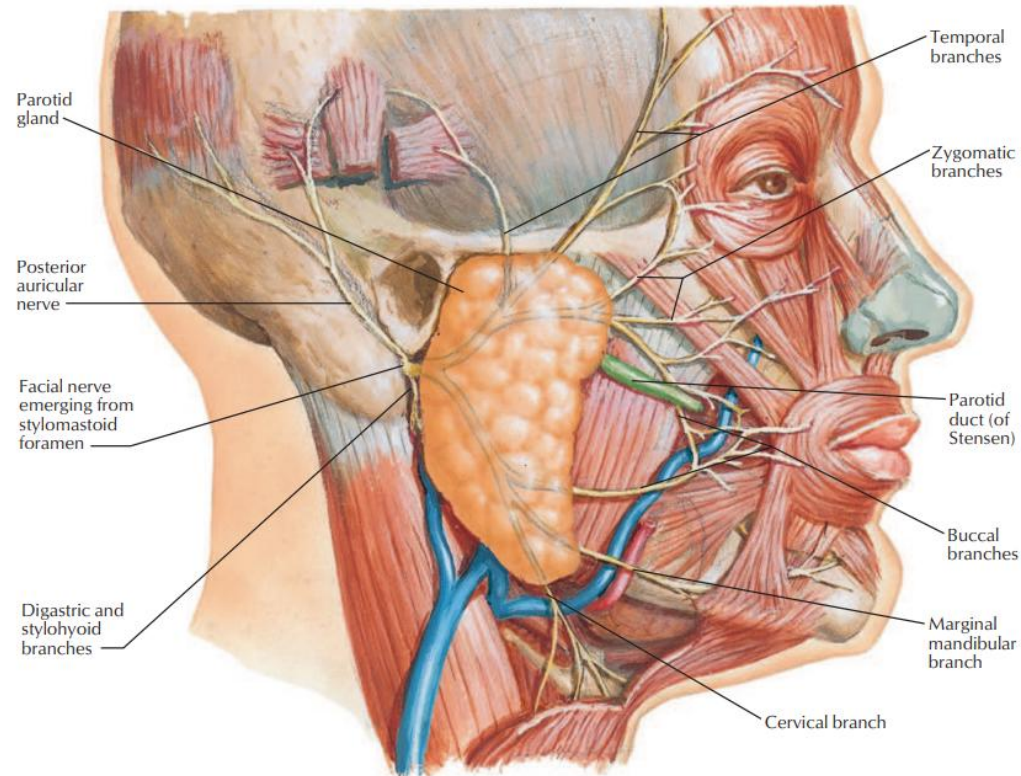
- Carries *pre-ganglionic parasympathetic fibers* to submandibular ganglion
- Carries special sensory (taste) fibers from the anterior 2/3 of the tongue back to the nucleus solitarius via geniculate ganglion

4. Motor branches (extracranial)

(After exiting stylomastoid foramen)

Facial nerve - extracranial course

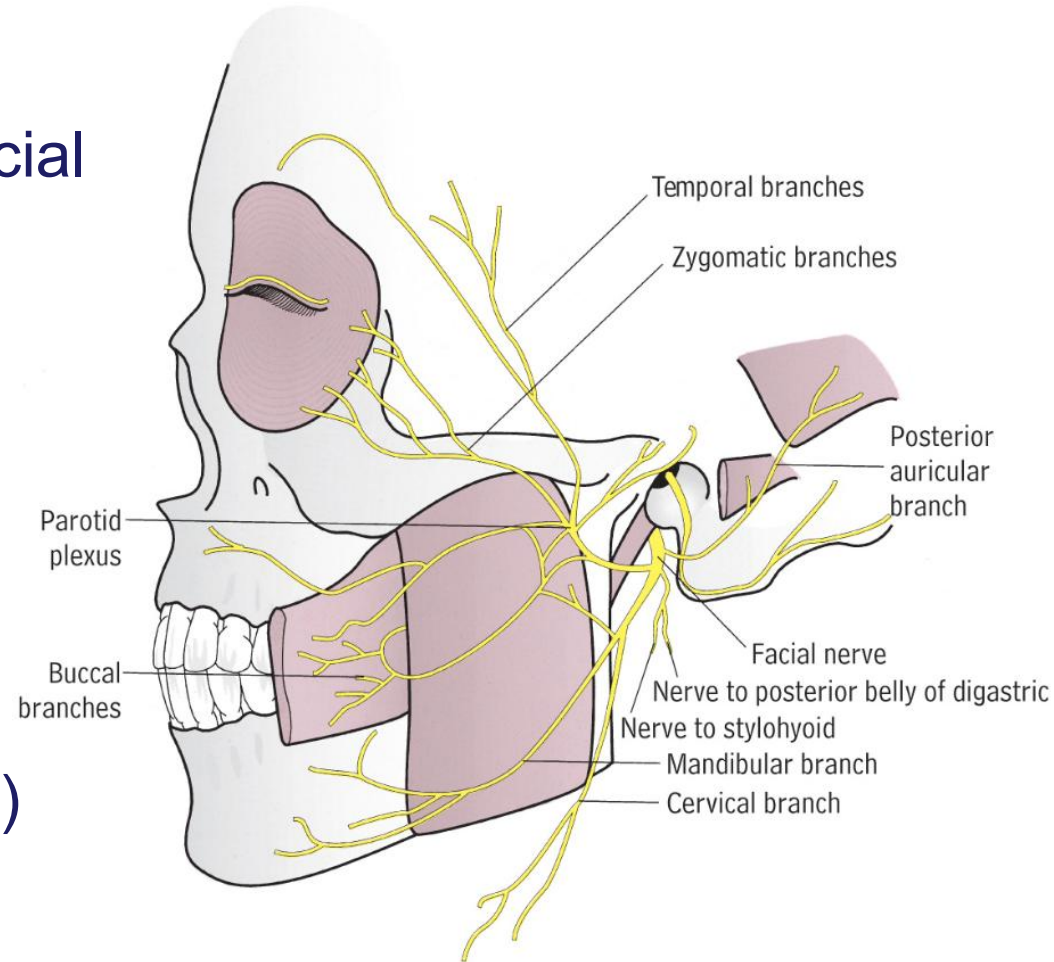
- Exits the **stylomastoid foramen**
 - motor only at this point
- Gives off **posterior auricular** and branches to the **posterior belly of digastric** and **stylohyoid**
- Passes anteriorly into the parotid gland
 - Divides into **two trunks** that pass through the parotid at different depths



Facial nerve - extracranial branches

Two trunks divide into 5 terminal branches that supply the muscles of facial expression:

1. Temporal
2. Zygomatic
3. Buccal
4. Mandibular (marginal)
5. Cervical



"Two Zebras Bit My Clavicle"

Facial nerve – lower motor lesions

Loss of functions at lesion:

#5

- loss of motor function to muscles of facial expression

#4

- loss of taste to anterior 2/3 of tongue
- loss of parasympathetics to submandibular and sublingual glands

#3

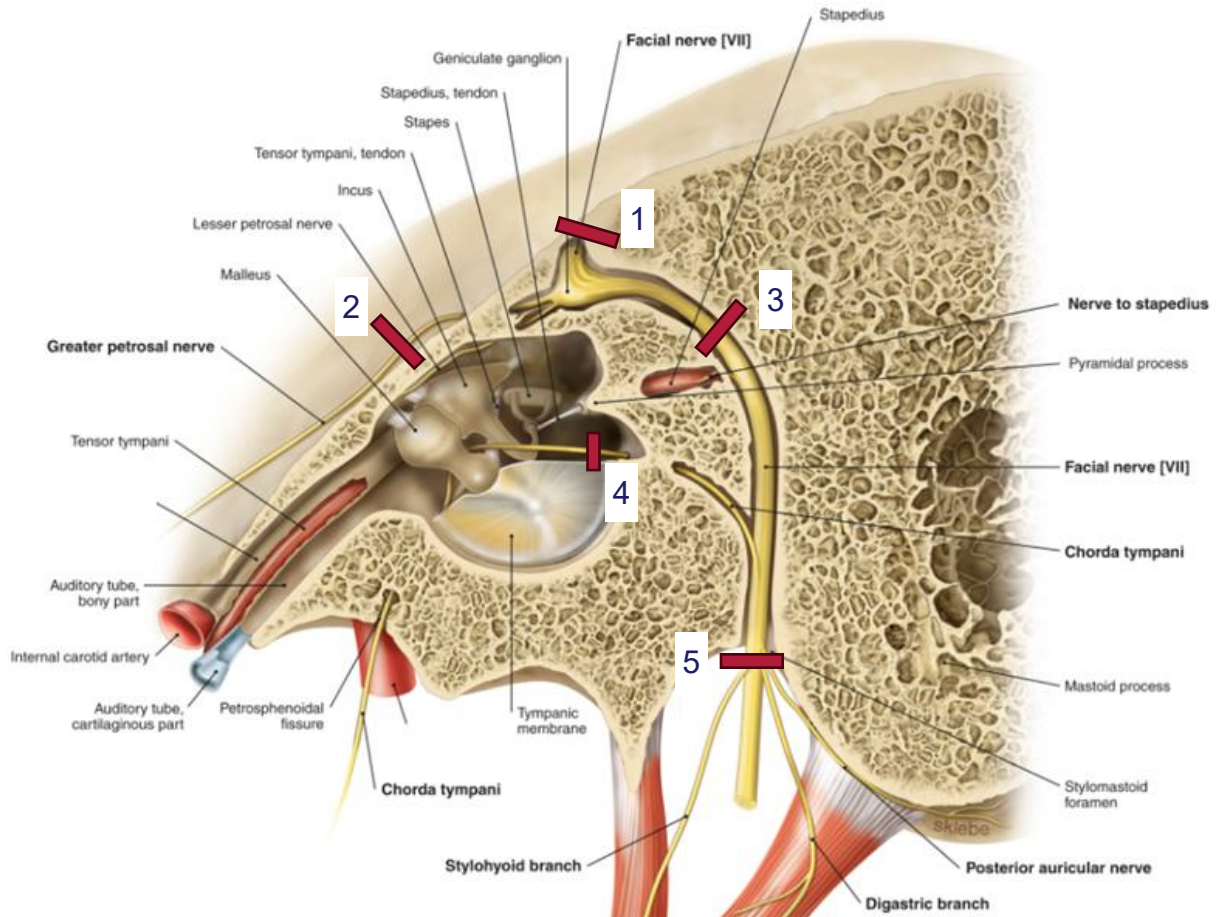
- loss of motor function
- loss of taste to anterior 2/3 of tongue
- loss of parasympathetics to submandibular and sublingual glands

#2

- loss of parasympathetics to nasal cavity, paranasal sinuses, lacrimal gland

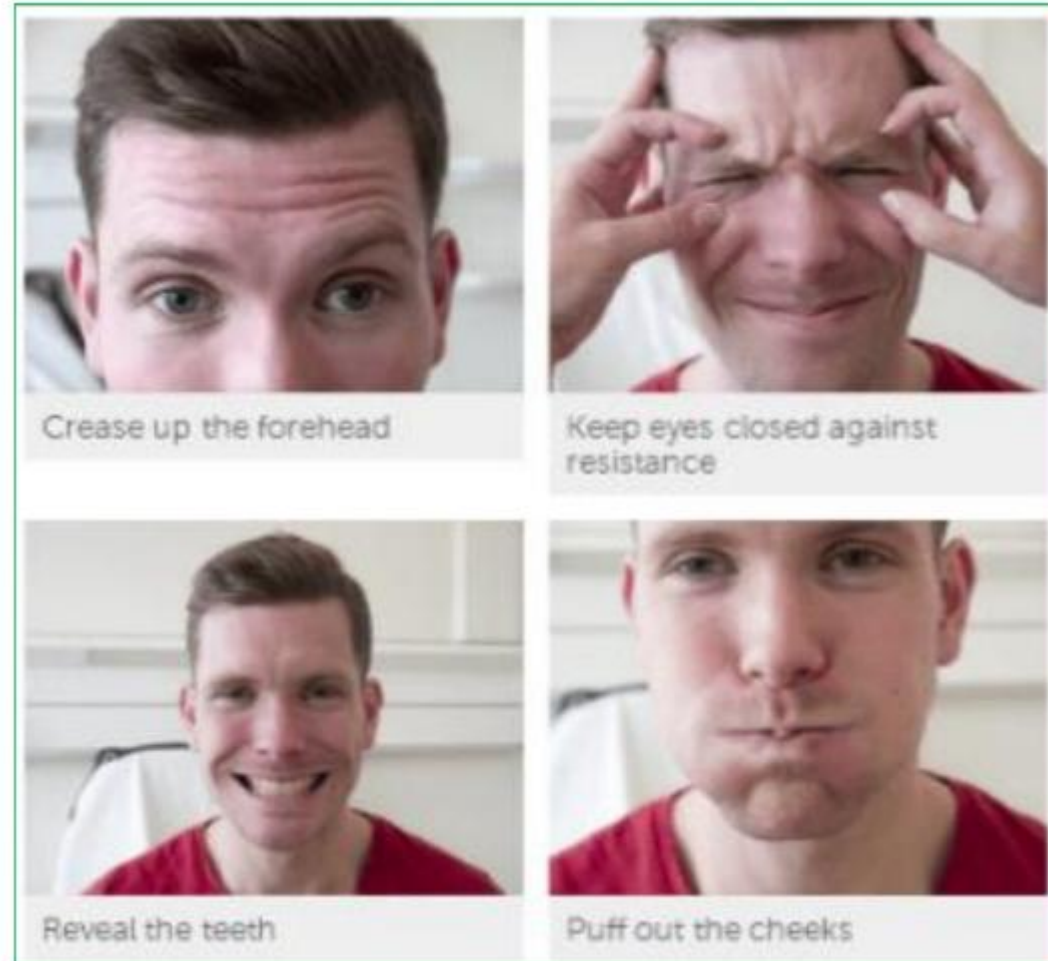
#1

- all functions lost



Clinical Testing of the facial nerve

- Crease the forehead
 - Frontalis muscle
 - Occipitalis muscle
- Keep eyes closed against resistance
 - Orbicularis oculi
- Reveal the teeth
 - Orbicularis oris
- Puff out the cheeks (or whistle)
 - Buccinator



Clinical Correlate: Bell's Palsy

Any injury that presents risk to the facial nerve may in paralysis of associated muscles

May be due to:

- infection
- trauma
- surgery
- tumors

What happens if the patient can frown?

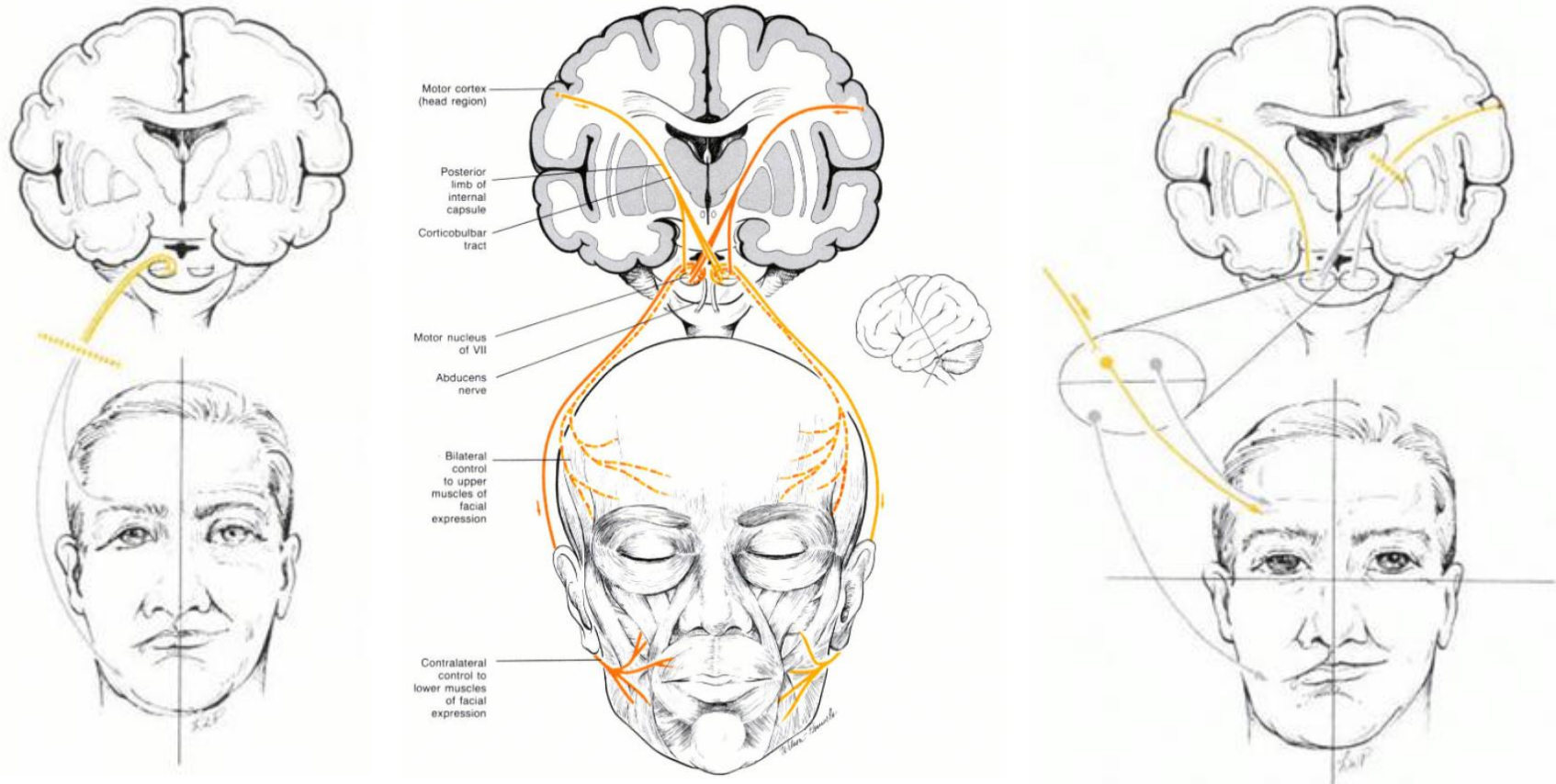
Left Peripheral VII Facial Weakness

Attempt to close eye results in eyeball rolling superiorly exposing sclera (Bell phenomenon) but no closure of the lid per se.



Patient unable to wrinkle forehead; eyelid droops very slightly; cannot show teeth at all on affected side in attempt to smile; and lower lip droops slightly.

Motor Neuron Lesions: Upper vs Lower



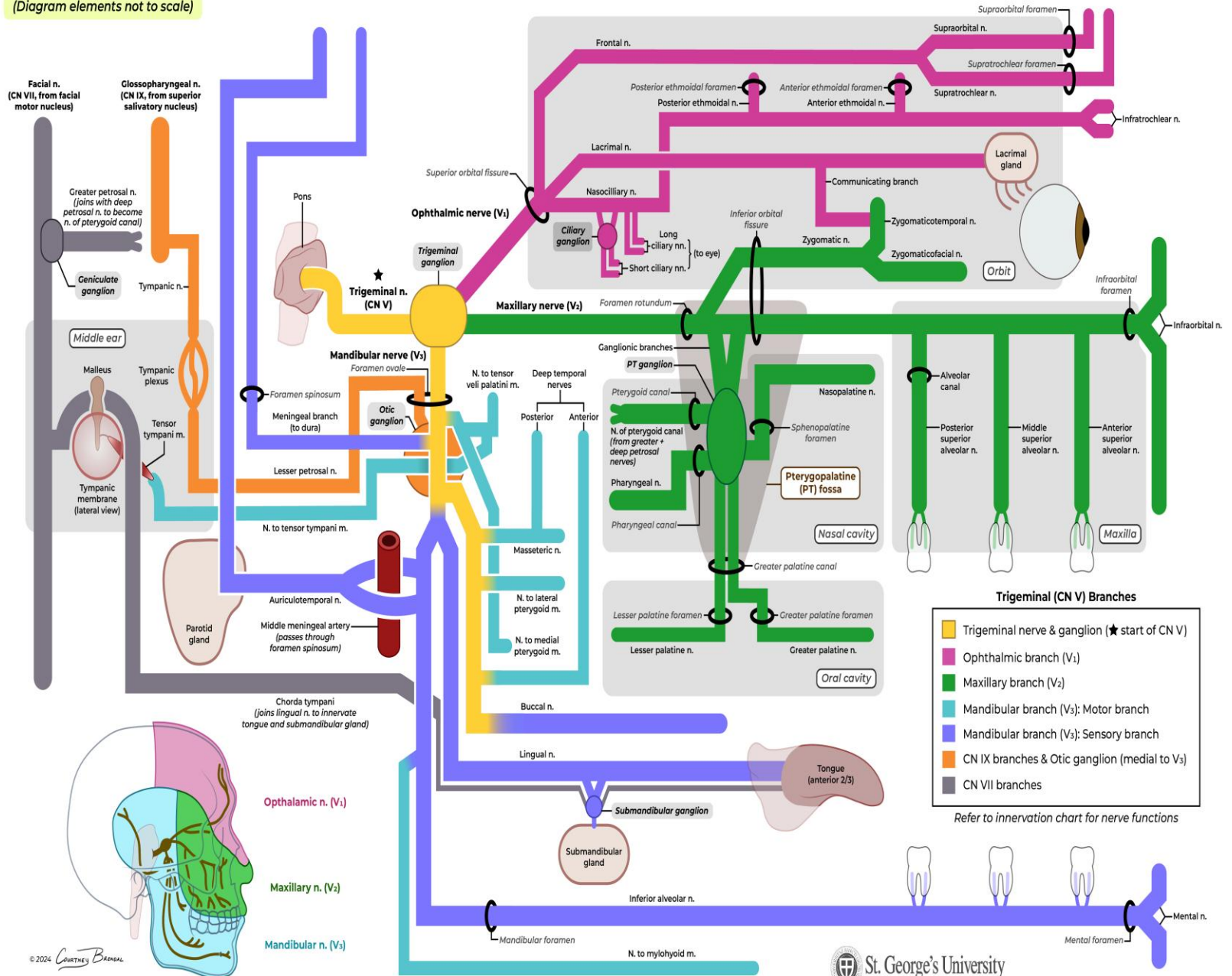
- LMN lesion

- i.e. within the brainstem or spinal cord
- Upper and lower halves of the face are affected
 - Complete palsy on one side of the face

- UMN lesion

- i.e. within the cortex
- Upper half of the face is spared ("upper spares upper")
- Lower half of the face is affected

(Diagram elements not to scale)

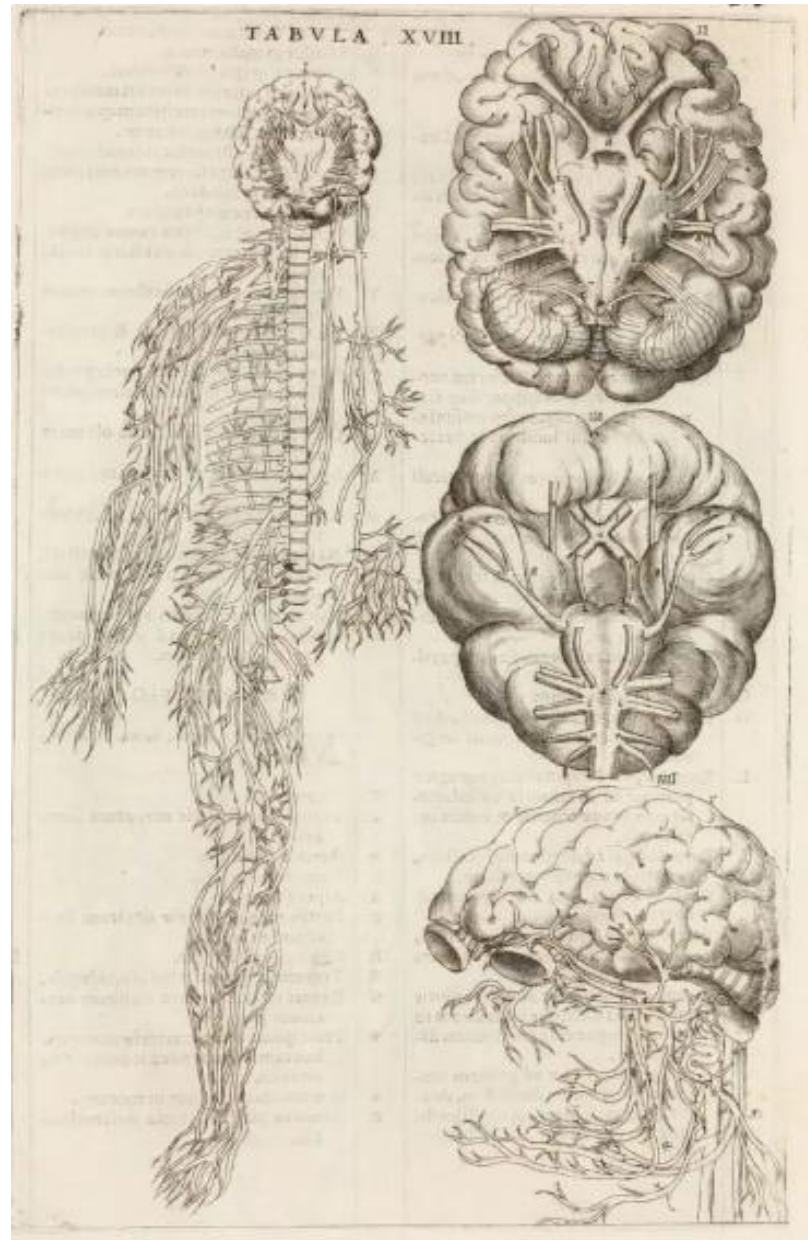


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Summary

	Trigeminal	Facial
Number	V	VII
Nuclei	3 sensory: • chief • mesencephalic • spinal 1 motor: • trigeminal	1 special sensory: • nucleus solitarius 1 motor: • facial 2 parasympathetic: • superior salivatory • lacrimal
Brainstem attachment	Lateral merging with middle cerebellar peduncle	Ventrolateral to the pontomedullary junction
Ganglia	Trigeminal in posterior portion of cavernous sinus	Geniculate in petrous temporal bone
Major branches	• Ophthalmic • Maxillary • Mandibular	• Greater petrosal • Chorda tympani

fin



Florentini Artis Medicinalis Tomus Tertius (circa 1611)
By Vidus Vidius (namesake of Vidian canal)